

GS-3 Compiled Database (Readymade Intros & Conclusions)

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INDIAN ECONOMY

Section Group: Industrial Dynamics & Reforms

Microtopic: Effects of Liberalisation on the economy; Changes in Industrial policy & their effects on industrial growth

Subtopic: FDI

Card 1: Ease of Doing Business & PLI

QUOTE

"From red tape to red carpet is a journey of mindset and law."

INTRODUCTION

- Improving the **ease of doing business** through reforms like the **Insolvency & Bankruptcy Code**, **PLI schemes**, and **faceless approvals** reflects the shift to a **"compliance-light, trust-heavy" economy**.

EXAMPLES

- IBC enabled recovery of **₹2.9 lakh crore** in major insolvency cases.
- PLI schemes across 14 sectors** attracted investment and scaled exports.
- MCA21, GSTN, and single-window clearances boosted **regulatory transparency**.

CONCLUSION

- India must deepen reforms in **contract enforcement and informal sector inclusion** for holistic ease.

Card 2: FDI/FII Trends & Trade Policies

QUOTE

"Capital follows clarity, consistency, and confidence."

INTRODUCTION

- India's evolving **FDI and trade policies**, coupled with improved ease of doing business, aim to harness **"trust reshoring"** and make India a **key node in global value chains**.

EXAMPLES

- India received **\$80+ billion FDI**, led by renewables, electronics, and services.
- Sectoral caps liberalised in **insurance (74%) and defence (74% automatic, 100% via govt route); 100% FDI permitted in space sector** under amended rules.
- India signed **FTA pacts with UAE and Australia** for export market access.

CONCLUSION

- A balance of **open markets and strategic safeguards** is vital for resilient industrial expansion.

Subtopic: LPG Reform and Impact

Card 1: Disinvestment Policy & Asset Monetization

QUOTE

"The government has no business to be in business." – NITI Aayog Vision Document

INTRODUCTION

- Strategic disinvestment and asset monetization help the state exit non-core sectors while unlocking capital. Through **"funding the future without expanding deficits,"** these tools enhance fiscal space.

EXAMPLES

- **IDBI Bank** saw progress on strategic disinvestment.
- NMP monetized **brownfield assets** in highways and power.
- **LIC IPO** broadened retail access to public wealth.

CONCLUSION

- A **transparent, time-bound disinvestment policy** is key to efficiency and capital mobilization.

Card 2: LPG Reforms

QUOTE

"Economic reforms begin where control ends."

INTRODUCTION

- The **LPG reforms transformed India** from a closed, socialist economy to a competitive market-led one. They laid the foundation for **India@100 as a developed nation**, promoting global integration and sectoral liberalisation.

EXAMPLES

- Post-reform period saw **GDP growth averaging above 6%** for decades.
- Industrial licensing abolished in **nearly all sectors**.
- **Services exports became a major growth driver**, led by IT and finance.

CONCLUSION

- Future progress must combine **reform continuity with inclusive safety nets** to widen the benefits of liberalisation.



Section Group: Macroeconomics & Fiscal Policy

Microtopic: Government Budgeting and fiscal management policies

Subtopic: Current Budget

Card 1: Budget Process & Components

QUOTE

"A budget is not just numbers on a page, it is an embodiment of our values." – Barack Obama

INTRODUCTION

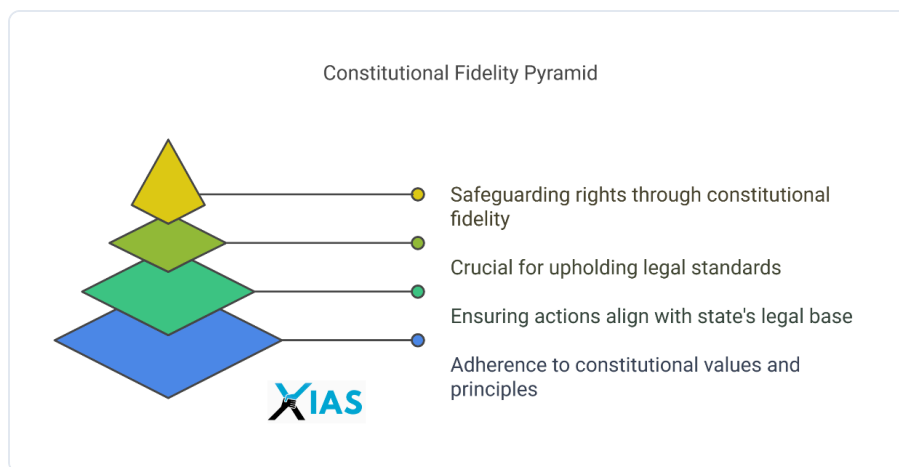
- India's budget process reflects both **constitutional fidelity (Article 112)** and developmental foresight, structured around estimates, receipts, and expenditures. Guided by the **FRBM Act**, it aims to ensure macroeconomic stability while **"planning prosperity through prudence."**

EXAMPLES

- Interim budget focused heavily on **revenue expenditure**.
- Fiscal deficit achieved **4.4% of GDP for FY26; FY27 target is 4.3% (CGA data)** due to high subsidies.
- 16th Finance Commission** asked to review **FRBM targets**.

CONCLUSION

- A **calibrated fiscal framework** with an **independent fiscal council** can balance growth imperatives with discipline.



Card 2: Outcome & Performance Budgeting

QUOTE

"You can't manage what you can't measure." – Peter Drucker

INTRODUCTION

- India's budgeting is evolving to measure impact, not just inputs—**"from outlay to outcome"**. By integrating **performance-based indicators**, it enhances transparency and public accountability.

EXAMPLES

- PMAY dashboard** tracks unit-wise completion status.
- Ministries like **Rural Development** linked funds to measurable outputs.
- NITI Aayog's Performance Index** shapes fund allocations.

CONCLUSION

- Mainstreaming **real-time tracking** and **performance-linked disbursement** can ensure outcomes match allocations.

Subtopic: Fiscal Policy

Card 1: Revenue & Capital Expenditure Trends

QUOTE

"Capital expenditure is the soul of sustainable growth."

INTRODUCTION

- India's fiscal strategy is undergoing a shift **from revenue-heavy budgets to capex-led growth engines**, in line with the aim of **"spending less and building more."** This approach enhances productivity, infrastructure, and private investment.

EXAMPLES

- Capex allocation increased to over **₹11 lakh crore** for infra.
- Fertilizer and food subsidies kept **revenue outgo** elevated.
- PM Gati Shakti got enhanced **capital infusion** for logistics.

CONCLUSION

- Capex focus must continue, coupled with **efficient monitoring** to ensure long-term returns and fiscal resilience.

Card 2: Government Schemes vs Capital Creation

QUOTE

"Schemes must not just spend, they must build."

INTRODUCTION

- Welfare schemes must increasingly focus on **productive asset creation**, aligning with the government's **"Minimum Government, Maximum Governance"** philosophy. Capital-linked welfare ensures both inclusion and economic expansion.

EXAMPLES

- Jal Jeevan Mission** built lasting rural infrastructure.
- PM Awas Yojana-Urban** spurred housing-linked capex.
- Rationalized **MGNREGS** outlay redirected toward durable assets.

CONCLUSION

- Linking welfare with **long-term capital assets** ensures **economic multipliers without fiscal stress**.



Card 3: Central & State Budget Interface

QUOTE

"Cooperative federalism must reflect in cooperative fiscalism."

INTRODUCTION

- The effectiveness of India's public finance depends on **centre-state fiscal synergy**, where **"collaborative spending powers"** ensure harmonized development. The Union budget now increasingly incentivizes state-level capital creation.

EXAMPLES

- **50-year interest-free loans** offered to states for infra projects.
- **GST compensation adjustments** aided states' fiscal balance.
- **PM Gati Shakti** involves coordinated planning across levels.

CONCLUSION

- A robust **intergovernmental fiscal framework** is vital for equitable, outcome-driven development across states.

Microtopic: Inclusive growth and issues arising from it

Subtopic: Inclusive Growth

Card 1: Inclusive Growth

QUOTE

"JAM is the highway to equity in a digital India." — Economic Survey

INTRODUCTION

- Inclusive growth **is driven by** JAM Trinity, Direct Benefit Transfers, **and** Jan Dhan–Aadhaar–Mobile **synergy**, making entitlements **portable, paperless, and person-centric**, aligned with **SDG 1 & 10**.

EXAMPLES

- **DBT** transferred ₹30 lakh crore+ across 400+ schemes directly into beneficiaries' accounts.
- **PM Jan Dhan Yojana** crossed 50 crore bank accounts milestone.
- **UPI** transactions surpassed ₹18 lakh crore monthly, promoting digital access.

CONCLUSION

- India's model of digital inclusion proves that **technology + trust = transformation**—a mantra for bridging developmental divides.

Subtopic: Inclusive Growth X Sustainable Development

Card 1: Role of SHGs, Cooperatives & Skill Dev

QUOTE

"Self-help is the best help—especially when it builds an economy from below." — Inspired by grassroots movements

INTRODUCTION

- Schemes like **MUDRA**, **PM-SVANidhi**, and **DAY-NRLM** empower **SHGs**, **nano-enterprises**, and informal vendors, becoming catalysts of "**Vocal for Local**" and "**Empower the last mile first**".

EXAMPLES

- **MUDRA loans** crossed ₹23 lakh crore mark, majority to women & SC/STs.
- **PM-SVANidhi** onboarded over 70 lakh street vendors with microcredit and e-KYC.
- **Skill India Mission** trained over 1.5 crore youth under PMKVY 4.0.

CONCLUSION

- People-led micro-entrepreneurship is India's answer to **inclusive capitalism**, unlocking **dignity, credit and capacity** at the grassroots.

Microtopic: Indian Economy and issues relating to planning, mobilization of resources, GDP,

and monetary framework

Subtopic: Investment

Card 1: Resource Mobilization

QUOTE

"Revenue is the lifeblood of a government, transparency its heartbeat." — Inspired by the Kelkar Committee

INTRODUCTION

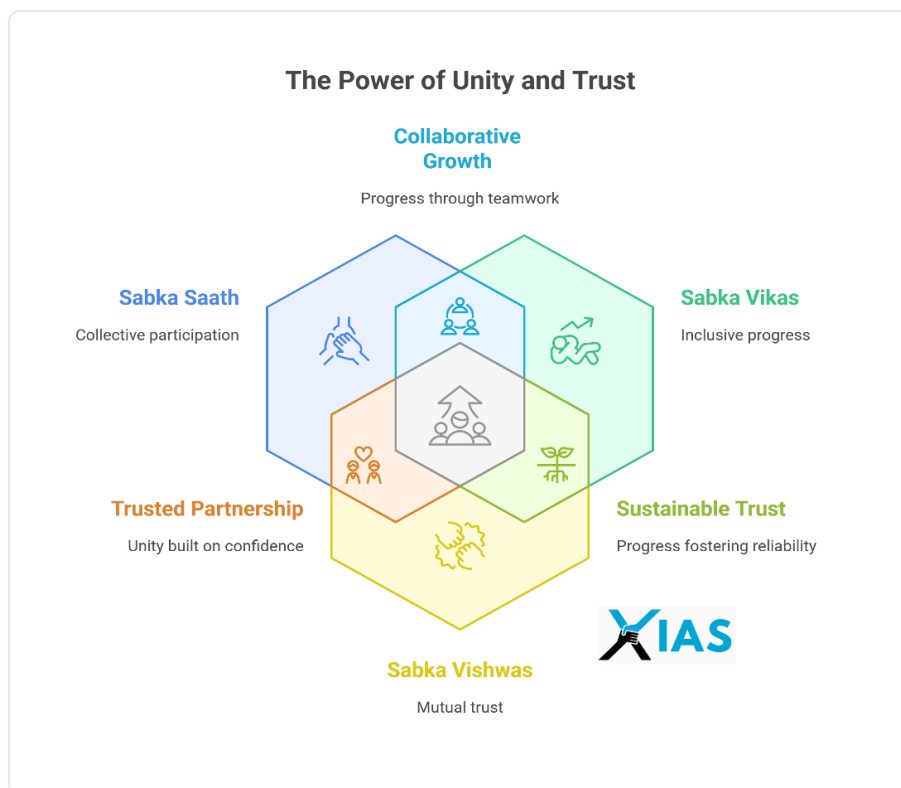
- With **Make in India**, **Invest India**, and **Digital India** as facilitators, India aims to boost **resource mobilisation** through **efficient taxation**, deepened **capital markets**, and stable **foreign investments**.

EXAMPLES

- Faceless tax regime** and Vivad Se Vishwas scheme improved tax compliance.
- SEBI** rolled out T+0 trade settlement to boost market liquidity.
- FDI** norms liberalised in **space**, **semiconductor**, and **green energy** sectors.

CONCLUSION

- A dynamic resource mobilisation strategy ensures **fiscal strength**, attracts global trust, and powers the engine of **inclusive and self-reliant growth**.



Subtopic: Niti Aayog X Planning Commission

Card 1: Planning & Economic Strategy

QUOTE

"A vision without a plan is just a wish." – Narendra Modi

INTRODUCTION

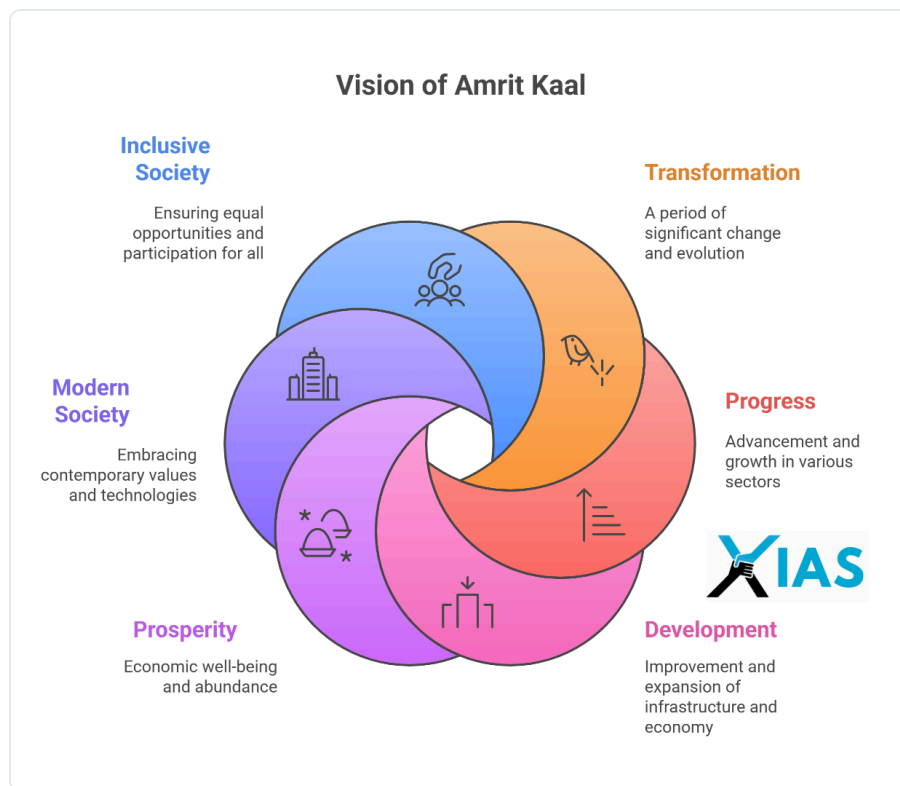
- India's journey from **centralised Five-Year Plans** to **Viksit Bharat@2047** reflects a shift towards **mission-mode governance** and **competitive federalism**, guided by **NITI Aayog's reform-driven think tank model** and "**Sabka Prayas**" philosophy.

EXAMPLES

- NITI Aayog** released sector-specific strategy papers aligned with Viksit Bharat goals.
- State governments were ranked under **SDG India Index**, promoting outcome-based planning.
- Finance Commission incentivised states using performance-linked grants.

CONCLUSION

- Strategic planning must align **vision with execution**, ensuring economic reforms are **federal, future-ready, and inclusive**, building a roadmap for **Amrit Kaal**.



Card 2: NITI Aayog Strategy & Viksit Bharat

QUOTE

"What India does today will shape the world of tomorrow." — Viksit Bharat Vision

INTRODUCTION

- Viksit Bharat@2047 **envisions** a prosperous, inclusive and climate-smart India, **guided by** NITI Aayog's **thematic roadmaps**, aligning national ambitions with **SDG 2030 and India@100** goals.

EXAMPLES

- Sectoral Viksit Bharat targets released for MSMEs, health, and innovation.
- NITI Aayog launched policy dialogues with states on export hubs and skilling.
- Paper on energy transitions laid roadmap for Net Zero by 2070.

CONCLUSION

- Strategic visioning must translate into **cohesive policy, cooperative execution, and catalytic change**—making India a global model of development by 2047.

Microtopic: Issues relating to overall economic growth, sector manufacturing, and employment generation

Subtopic: Economic Growth

Card 1: Growth vs Development

QUOTE

"Growth is hollow if it doesn't bridge the divide." — Inspired by the Human Development Report

INTRODUCTION

- India's development narrative balances **GDP growth** with improvement in **HDI, GDI, and Multidimensional Poverty Index**, reflecting the push toward "**Sabka Saath, Sabka Vikas, Sabka Vishwas**".

EXAMPLES

- NITI Aayog released updated **MPI report** identifying high-deprivation districts.
- Government focused on **Aspirational Districts Programme** for equitable outcomes.
- UNDP data showed progress in **life expectancy and schooling years**.

CONCLUSION

- India must prioritise **inclusive indicators** alongside GDP to ensure growth transforms into **real, measurable and people-centric development**.

Subtopic: Manufacturing Sector

Card 1: Make in India & Start-Up India

QUOTE

"India must become the factory of the world without losing its soul."

INTRODUCTION

- Schemes like **Make in India**, **Start-Up India**, and **industrial corridors** aim to boost domestic manufacturing, jobs, and innovation by building a **"production-driven, globally integrated India."**

EXAMPLES

- Over **1.4 lakh startups** recognized, creating **employment across 700+ districts**.
- Mobile manufacturing exports** crossed ₹1.2 lakh crore.
- DMIC and other corridors** operationalised with smart infrastructure nodes.

CONCLUSION

- A push for **design-led, tech-integrated industrialisation** will future-proof India's production base.

Card 2: MSME Sector Reforms

QUOTE

"Small businesses build big economies."

INTRODUCTION

- The **MSME sector**, employing over **11 crore people**, is pivotal for inclusive industrial growth. Initiatives like Emergency Credit Line, Udyam portal, and Credit Guarantee Schemes **aim to enable** "formal strength from informal roots."

EXAMPLES

- ₹3.6 lakh crore disbursed under **ECLGS** to 1.2 crore MSMEs.
- Udyam Assist Platform** onboarded over 2 crore micro enterprises.
- SIDBI launched **digitised loan pipelines** for small-scale units.

CONCLUSION

- Targeted credit, **digital skilling**, and **cluster competitiveness** can unlock MSME-led growth momentum.

Subtopic: Unemployment

Card 1: Employment

QUOTE

"Employment is not just about income—it's about identity and dignity." — World Bank

INTRODUCTION

- In the world's youngest nation, addressing **structural unemployment** and formalising the **gig and informal economy** is key to leveraging the **demographic dividend** and building "**Atmanirbhar Bharat**".

EXAMPLES

- e-Shram** portal enrolled over 29 crore unorganised workers.
- NITI Aayog's gig report** recommended social security for platform workers.
- PLFS 2025 Annual Report** shows stable overall LFPR at **59.3%**, with female LFPR significantly improved to **40.0%**. Overall unemployment rate declined to **3.1%**.

CONCLUSION

- India's employment strategy must go beyond numbers to ensure **dignified, secure and future-ready livelihoods** across sectors and skill levels.

Section Group: Physical Infrastructure & Capital

Microtopic: Capital investment models and public-private frameworks

Subtopic: PPP X Infrastructure

Card 1: Investment Models

QUOTE

"Public-private synergy builds more than infrastructure—it builds trust."

INTRODUCTION

- India's infra push is increasingly led by hybrid models like PPP, HAM, BOT, and EPC, **helping mitigate financial risk**. These models reflect the **philosophy of "collaborative growth through shared responsibility."**

EXAMPLES

- Bharatmala highways** built using HAM and EPC mix.
- Solar parks** leveraged PPP with Viability Gap Funding.
- Vande Bharat** trains procured under innovative PPP format.

CONCLUSION

- Improving **risk-sharing frameworks** and **speedy dispute resolution** will rejuvenate private investment in infra.

Microtopic: Infrastructure systems connectivity: energy, ports, roads, airports, and railways

Subtopic: Airports X Infrastructure

Card 1: Railways Modernisation & Airports

QUOTE

"Speed and sustainability are the twin engines of modern mobility."

INTRODUCTION

- Rail and air networks are being reimagined under **Vande Bharat**, **UDAN**, and the **Amrit Bharat Station Scheme**, embodying the **Amrit Kaal infrastructure push** for **fast, green and inclusive mobility solutions**.

EXAMPLES

- 60+ Vande Bharat trains** deployed on major routes.
- 500+ stations** under Amrit Bharat redevelopment plan.
- 480+ air routes** opened under UDAN.

CONCLUSION

- Smart design and affordability** should guide the mobility transition across Bharat and India.



Subtopic: Renewable Energy

Card 1: Energy Infrastructure

QUOTE

"Clean energy isn't a luxury, it's a national imperative."

INTRODUCTION

- India is transforming into a **Green Growth Leader** by investing in renewables, battery storage, and **green hydrogen**, aligning infrastructure with **Panchamrit climate targets** and a **Net-Zero 2070 vision**.

EXAMPLES

- Solar capacity surpassed **80 GW**, among the world's top 5.
- Green Hydrogen Mission targets **5 MMT production with ₹20,000+ crore** support.
- Hybrid RE parks** launched in Gujarat and Rajasthan.

CONCLUSION

- Stable policy, **storage infrastructure**, and green finance are key to achieving clean energy security.

Card 2: ENERGY — General Overview

INTRODUCTION

- India is the **3rd largest energy consumer** globally.
- Highest growth rate** of energy consumption — **11%**.
- Energy Trilemma: Energy equity + Energy security + Sustainable energy**.

DATA

- Since 1850s — energy use has become **25×**.

Card 3: NATURAL GAS

INTRODUCTION

- Target remains **15% of the energy mix by 2030**.
- Cleanest among fossil fuels**.
- Currently, natural gas constitutes around **3.9% of utility capacity**, with nearly **45–50% of requirements met via LNG imports**.
- TAPI Pipeline** — key infrastructure project.

DATA

- Currently, natural gas constitutes around **3.9% of utility capacity**, with nearly **45–50% of requirements met via LNG imports**.

Card 4: GEOTHERMAL ENERGY

INTRODUCTION

- **Definition:** Energy generated and stored inside Earth's crust.
- **1st ever geothermal field development** in India — **Puga Valley, Ladakh.**

DATA

- India has **350 sites** with potential of **10 GW**.

Card 5: METHANOL ECONOMY

INTRODUCTION

- **Methanol economy** = replacing dependence on oil and coal with **methanol**.
- Slightly **lower energy content** than petrol and diesel.
- **NITI Aayog Roadmap:** Substitute **10% of crude imports by 2030** through methanol alone.

DATA

- Target: **10% substitution of crude imports by 2030.**

Card 6: BIO-ETHANOL

INTRODUCTION

- India successfully achieved its **E20 target** ahead of schedule, with a national average blending rate of **~20% in the 2025–26 Ethanol Supply Year (ESY)**.
- **PM JIVAN scheme** — support for 2G bio-ethanol.
- Government should incentivise **flex-fuel engines** in vehicles.

DATA

- India successfully achieved its **E20 target** ahead of schedule, with a national average blending rate of **~20% in the 2025–26 Ethanol Supply Year (ESY)**. Total ethanol production capacity has expanded to nearly **20 billion litres**.

Card 7: PM KUSUM

INTRODUCTION

- Target: Add **30.8 GW solar capacity** + **de-dieselisation** + feeder solarisation.

DATA

- **30.8 GW** solar capacity target.

Card 8: NATIONAL GREEN HYDROGEN MISSION

INTRODUCTION

- **Definition:** Producing H₂ from water using only **green & renewable sources**.
- **Types:**
 - 1. **Black/Brown** (coal).
 - 2. **Grey** (natural gas).
 - 3. **Blue** (carbon capture).
 - 4. **Green** (renewable).
- Centre cleared **National Green Hydrogen Mission** of **₹19,000 crore**.
- **Goals by 2030:**
 - 1. **5 MMT** of green hydrogen.
 - 2. **SIGHT** — Strategic Interventions for Green Hydrogen Transition Programme (financial incentives for electrolyzers).
- **Advantages:**
 - 1. Reduce fossil fuel imports by **₹1 lakh crore**.
 - 2. Decarbonise industry/mobility/energy.
 - 3. Global hub for derivatives.
- **Issues:**
 - 1. India <1% of global production.
 - 2. Immature tech.
 - 3. **3× more expensive than Black H₂** — CEEW.

CONCLUSION

- Game changer in reshaping energy landscape, **mitigating climate change** while fuelling economic progress.

DATA

- Mission budget: **₹19,000 crore**.
- Target: **5 MMT** by 2030.
- Currently India: **<1%** of global production.

Subtopic: Transport

Card 1: Transport: Bharatmala, Sagarmala, Gati Shakti

QUOTE

"Infrastructure is the foundation stone of a nation's economic destiny."

INTRODUCTION

- Through **Bharatmala**, **Sagarmala**, and the **PM Gati Shakti National Master Plan**, India is creating unified transport systems and reducing logistics bottlenecks—marking a decisive shift toward a **"One Nation, One Infrastructure"** paradigm for integrated growth.

EXAMPLES

- 35+ logistics parks** approved under Bharatmala Phase-I.
- Gati Shakti portal** digitally monitors 2,000+ infra projects.
- Sagarmala enabled **port-linked clusters** in coastal regions.

CONCLUSION

- Accelerating **multimodal infrastructure with digital governance** will unlock seamless connectivity and regional equity.

Card 2: Urban Infra: Smart Cities & Housing

QUOTE

"Well-planned cities are not built overnight—they are coded into policy and design."

INTRODUCTION

- Urban renewal through **Smart Cities Mission**, **AMRUT 2.0**, and **PMAY-Urban** reflects India's effort to build climate-resilient, tech-enabled and inclusive cities, **consistent with the aspiration of India@100** as a **sustainable urban civilisation**.

EXAMPLES

- 7,000+ smart infrastructure projects** executed under Smart Cities.
- 500+ cities** achieving water tap coverage via AMRUT.
- Over **1.3 crore houses sanctioned** under PMAY-U.

CONCLUSION

- A citizen-first approach and **climate-smart design** will shape the next generation of Indian cities.

Card 3: Logistics & Digital Infra

QUOTE

"Connectivity is productivity in the Bharat of today."

INTRODUCTION

- From **PMGSY rural roads** to **Digital India fibre backbones** and the **National Logistics Policy**, India is building last-mile assets that reduce costs and bridge the rural-urban divide—driving the vision of a **“Viksit Bharat powered by seamless access.”**

EXAMPLES

- PMGSY built over **7.5 lakh km** of all-weather rural roads.
- 2.1 lakh Gram Panchayats** connected via BharatNet.
- The recent **DPIIT-NCAER report (September 2025)** assesses that India's logistics cost has dropped to **7.97% of its GDP**, officially placing it in the single-digit category alongside advanced economies.

CONCLUSION

- Synchronising **physical and digital infrastructure** is crucial for inclusive economic transformation.

AGRICULTURE

Section Group: Agriculture & Farm Dynamics

Microtopic: Irrigation design, management, water efficiency, and conservation infrastructure

Subtopic: Irrigation system

Card 1: Irrigation & Watershed Development

QUOTE

"More crop per drop is not just a slogan, it's a necessity."

INTRODUCTION

- Efficient irrigation is the fulcrum of Indian agriculture. With only about **50% of gross cropped area irrigated**, schemes like **PMKSY** and **watershed development** aim to bridge the **"irrigation inclusion gap."**

EXAMPLES

- Micro-irrigation coverage** expanded under PMKSY-Per Drop More Crop.
- Watershed Development Component aided **drought-proofing** in Bundelkhand.
- Solar-powered pumps** introduced under KUSUM for off-grid irrigation.

CONCLUSION

- Scaling up **micro and climate-resilient irrigation** must be priority, with **tech-driven water budgeting** at the local level.

Subtopic: Water Conservation

Card 1: WATERSHED DEVELOPMENT

INTRODUCTION

- **Definition:** Conservation, regeneration and judicious use of all natural resources in the watershed.
- **Schemes:**
 - 1. MGNREGA.
 - 2. Neeranchal.
 - 3. PMKSY.
 - 4. Jal Shakti Abhiyan.
- **Best Practices:**
 - 1. Hiware Bazar.
 - 2. Ralegaon Siddhi.
 - 3. Jal Swavalamban (RJ).
 - 4. Jalgaon model of Pani Panchayat.
- **Integrated Watershed Management:** NGOs — Pani Foundation, Sujala (KN).
- **Micro-watershed Atlas of India 2019.**
- **Mihir Shah Committee:**
 - 1. Demand mgmt — crop diversification.
 - 2. RRR.
 - 1. Supply side — nature-based solutions.
 - 2. Dam & river mgmt.
 - Water governance — multi-stakeholder **National Water Commission.**

CONCLUSION

- Way forward: **Agroforestry, ZBNF, sub-surface irrigation.**

Microtopic: Issues related to direct and indirect farm subsidies and minimum support prices

(MSP)**Subtopic: MSP****Card 1: MSP & Farm Subsidies****QUOTE**

"MSP is not a price, it's a promise of protection."

INTRODUCTION

- Input subsidies and MSP offer a **safety net for farmers**, yet their **fiscal sustainability and crop bias** raise concerns. Reforms must ensure **“support without distortion.”**

EXAMPLES

- MSP declared for 23 crops, but **procurement skewed toward paddy-wheat**.
- **PM-KISAN** transferred direct cash support to over 11 crore farmers.
- Fertilizer subsidy revised to **nutrient-based model** with DBT pilot.

CONCLUSION

- Subsidies must evolve into **targeted, direct support**, linked to **crop diversification and soil health**.

Microtopic: Major Crops, cropping patterns, and regional agriculture production styles

Subtopic: Cropping pattern

Card 1: Major Crops & Cropping Patterns

QUOTE

"Agriculture is not just a means of livelihood, it's the backbone of Bharat."

INTRODUCTION

- India's cropping patterns reflect its **diverse** agro-climatic zones, **but are increasingly misaligned with water and soil sustainability**. A shift toward "agro-ecological planning for **nutritional security**" is now crucial.

EXAMPLES

- Paddy area in Punjab reduced, replaced by **maize and pulses** under diversification schemes.
- Millet cultivation** surged in dry zones following the **International Year of Millets** push.
- Oilseed promotion** expanded in arid zones through targeted cluster development.

CONCLUSION

- Cropping choices must align with **climate, soil, and market linkages** to ensure resilient and sustainable agriculture.



Subtopic: Organic Farming

Card 1: ORGANIC FARMING

INTRODUCTION

- **Definition:** Sustainable agricultural approach avoiding synthetic inputs, emphasising **soil health, biodiversity and natural processes**.
- **Govt steps:**
 - 1. **PGS-India certification.**
 - 2. **Paramparagat Krishi Vikas Yojana (PKVY).**
 - 3. **NMSA.**
- Awareness campaign similar to **SBM** recommended for "**Swachh Food**".

DATA

- 1. India ranks **8th** in world's organic agricultural land.
- 2. **1st** in total number of producers (2020).
- Export of organic food products rose by **51%** in value.
- Only **2%** of the **140 mn ha** net sown area under organic farming.

Microtopic: Public Distribution System (PDS) management, buffer stock dynamics, food security missions, and economics of animal rearing

Subtopic: Buffer stock & food security

Card 1: SUSTAINABLE FOOD SYSTEMS

INTRODUCTION

- **Definition:** A food system that delivers food security and nutrition for all while being:
- **Economically sustainable** — profitable.
- **Socially sustainable** — broad-based benefits for society.
- **Environmentally sustainable** — positive or neutral impact.
- **Efforts:**
 - 1. **NMSA.**
 - 2. **Per Drop More Crop.**
 - 3. **Paramparagat Krishi Vikas Yojana.**

CONCLUSION

- **Precision agri + agro-climatic agri + adoption of tech.**

Subtopic: Non-Farm Activities (Livestock)

Card 1: Animal Rearing & Livestock Economics

QUOTE

"The strength of rural India lies beyond crops—in hooves, fins, and wings."

INTRODUCTION

- Animal husbandry, contributing over **30% to agri-GVA**, is key for **income diversification and nutritional security**. The focus is now on building **"blue, white, and pink revolutions 2.0."**

EXAMPLES

- Dairy cooperatives** digitally modernized under Rashtriya Gokul Mission.
- PM Matsya Sampada Yojana** scaled fishery clusters in coastal belts.
- Livestock insurance cover expanded in high-risk states through **NLM**.

CONCLUSION

- Strengthening **value chains, veterinary access**, and insurance is vital for a **vibrant allied agri-economy**.

Subtopic: PDS

Card 1: Food Security & PDS

QUOTE

"No one should go hungry in the land that grows food."

INTRODUCTION

- India's food security architecture—centered on **NFSA, PDS, and buffer norms**—ensures inclusive access. Yet, leakages and overstocking challenge the vision of **"hunger-free, waste-free Bharat."**

EXAMPLES

- One Nation One Ration Card** achieved full portability across states.
- PDS digitization reduced ghost beneficiaries in high-leakage districts.
- Buffer stock norms recalibrated for seasonal and climate shocks.

CONCLUSION

- Reforming PDS via **tech, Aadhaar, and community audit** can ensure efficient, demand-based food security.

Microtopic: Storage, transport, and marketing of agricultural produce along with digital e-

technology interventions

Subtopic: Agri-Marketing

Card 1: Agri Marketing & APMC Reforms

QUOTE

"Farmer incomes rise not just from what they grow, but how and where they sell."

INTRODUCTION

- Agricultural marketing reforms are vital to move from **price support to price realization**. Initiatives like **e-NAM**, **Model APMC Acts**, and **GrAMs** aim to create a **"One Nation, One Market"** ecosystem.

EXAMPLES

- Over **1,400 mandis** integrated with e-NAM to enable inter-state trade.
- States like **Madhya Pradesh** adopted APMC Act reforms for private markets.
- **GrAMs** operationalized with digital kiosks and warehousing linkages.

CONCLUSION

- India needs a **unified agri-market architecture** with robust logistics, digital access, and transparent pricing mechanisms.

Card 2: Post-Harvest Infrastructure & Cold Chains

QUOTE

"A farm's true value is realized only after the harvest."

INTRODUCTION

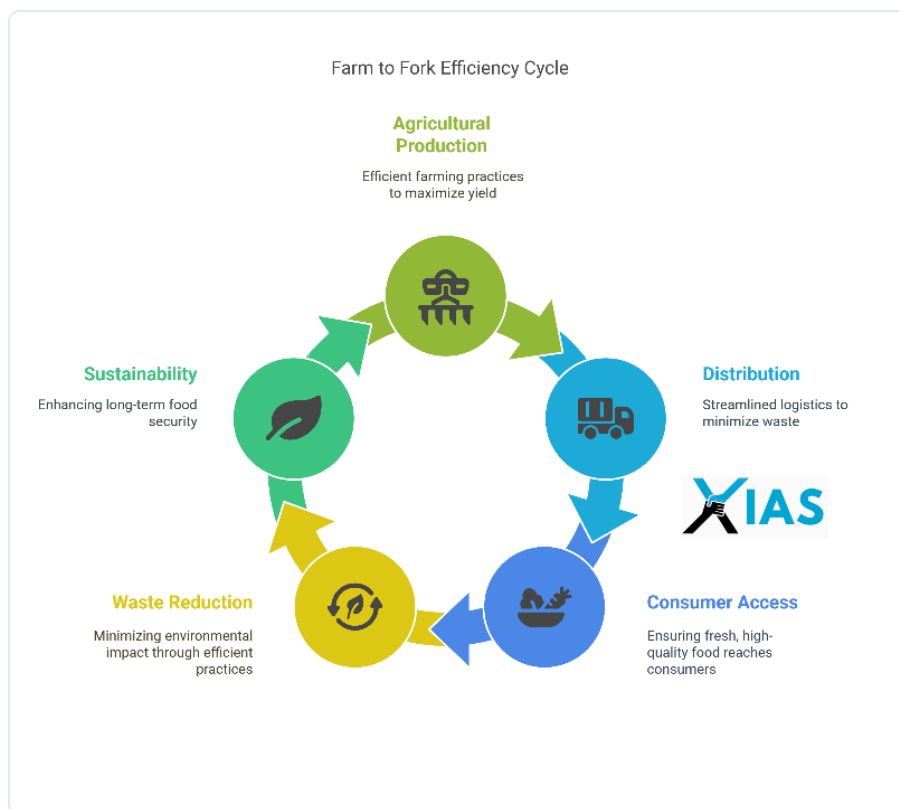
- India loses nearly **15–20% of its agri produce** post-harvest, **due to gaps in cold chains and logistics**. **Addressing this is key to achieving "farm to fork efficiency."**

EXAMPLES

- Mega Food Parks** received fresh capital under PMFME for cold storage and processing.
- ONDC integration** piloted for direct farm- to-retail logistics.
- NABARD supported **warehouse projects** through agri infra funds.

CONCLUSION

- Post-harvest infrastructure must be **localized, tech-integrated**, and aligned with farmer producer organizations.



Card 3: FPOs & Agri Startups

QUOTE

"From marginal farmers to market makers—FPOs are rural India's startups."

INTRODUCTION

- FPOs and agri startups empower smallholders by enhancing bargaining power, market access, and scale. They reflect the shift toward **"cooperative enterprise for competitive advantage."**

EXAMPLES

- Over **10,000 FPOs registered** under the Central Sector Scheme.
- Startups like **Agrowave** and **DeHaat** built end-to-end digital platforms.
- FPOs in Bihar and MP entered **contract farming for organic exports**.

CONCLUSION

- A supportive policy and digital ecosystem can make FPOs and startups the **backbone of rural industrial revival**.

Subtopic: Tech in Agri Production and Marketing

Card 1: Agri-Tech & Digital Farming

QUOTE

"From bullock carts to drones—Indian farming is flying into the future."

INTRODUCTION

- Emerging technologies like **Kisan drones, AI, and precision farming** are driving the transition toward **"smart agriculture."** These tools enhance productivity, reduce input misuse, and empower farmers.

EXAMPLES

- Kisan Drones** used in over 80,000 hectares for pesticide spraying.
- Agri-stack pilots launched for **farm-level data-driven advisories**.
- AI-based soil sensors deployed in clusters under **Digital Agriculture Mission**.

CONCLUSION

- Scaling **affordable, vernacular, and accessible agri-tech** is essential for inclusive, future-ready farming.

Subtopic: Tech in Farm Productivity and Income

Card 1: AGRITECH

INTRODUCTION

- MMS:** Agriculture suffers from **"technological fatigue"**.
- Goal: **Not just growing more but growing smarter & more sustainably**.
- Applications: refer all individual tech cards above.

Section Group: Food Processing Industry

Microtopic: Food processing and related industries (scope, significance, structural locations, and supply chain management)

Subtopic: Food Processing Sector

Card 1: Food Processing in Rural Economy

QUOTE

"Turning grain into gain—food processing multiplies farm value."

INTRODUCTION

- Food processing is the **missing link between farms and formal markets**, with the potential to double farmer incomes, generate rural jobs, and curb wastage. It strengthens the **"value from the village"** model through local entrepreneurship.

EXAMPLES

- Food processing sector contributes over **10% to manufacturing GVA**.
- Mega Food Parks created **over 60,000 rural jobs** directly.
- Rise in demand for **processed millets and organic produce** boosted rural enterprise.

CONCLUSION

- Unlocking the sector's potential requires **rural-centric industrial policy** and **skill-linked infrastructure**.

Card 2: Pradhan Mantri Kisan SAMPADA Yojana

QUOTE

"From field to factory to fork—SAMPADA builds the bridge."

INTRODUCTION

- The **PM Kisan SAMPADA Yojana** aims to build a robust processing ecosystem by promoting **value addition, cluster-based growth**, and reducing agri-waste. It reflects the spirit of **"processing-led prosperity."**

EXAMPLES

- Financial assistance extended to **over 900 processing units** across states.
- Agro Processing Clusters approved in **Maharashtra and Assam**.
- Waste reduction of up to **10–15%** reported in targeted zones.

CONCLUSION

- To scale impact, the scheme must focus on **ease of credit, backward linkages**, and export integration.

Card 3: Supply Chain Management & Logistics

QUOTE

"A strong supply chain is the artery of agro-industrial India."

INTRODUCTION

- Efficient logistics—from farm gate to processing hub—is essential to minimize spoilage and ensure quality. **"From surplus to seamless"** must be the approach in managing agri supply chains.

EXAMPLES

- ONDC integrated farmer produce** with retail logistics networks.
- Railways launched **Kisan Rails** for faster perishables transport.
- NABARD-funded logistics hubs supported FPO-led aggregation.

CONCLUSION

- Integrated **cold-logistics and multimodal supply networks** are critical to powering rural industrialization.

Card 4: Upstream & Downstream Linkages

QUOTE

"The strength of a food economy lies in its entire chain, not just one link."

INTRODUCTION

- Effective upstream and downstream linkages connect **producers to processors and processors to markets**. Such connections foster a **"farm-to-factory-to-market"** continuum that multiplies rural incomes.

EXAMPLES

- Agri export hubs** created downstream tie-ups with global retailers.
- Farmer groups linked to **Mega Food Parks** via primary collection centres.
- Startups used blockchain for **traceable supply chain assurance**.

CONCLUSION

- Strengthening **last-mile aggregation and first-mile distribution** is vital to a competitive rural value chain.

Card 5: Challenges in Food Processing Sector

QUOTE

"An unprocessed crop is an opportunity lost."

INTRODUCTION

- Cold chain gaps, complex licensing, and weak finance access limit rural processing. Tackling these hurdles is key to enabling a **"farm-to-fork without friction"** ecosystem.

EXAMPLES

- FSSAI eased **compliance norms** for micro food enterprises.
- Cold chain availability still less than **15% of demand** in perishables.
- Credit support under **CGTMSE** expanded for agri-processing units.

CONCLUSION

- Streamlined regulations, **customized rural finance**, and targeted infra support can unlock sectoral growth.

Section Group: Land Reforms

Microtopic: Land Reforms in India and legislative frameworks

Subtopic: Contract farming & Land leasing

Card 1: Land Leasing Reforms

QUOTE

"Unrecorded tenancies breed insecurity; formal leasing fosters productivity."

INTRODUCTION

- Modern land leasing frameworks aim to protect both landowners and cultivators, promoting **"ownership without alienation."** States are moving toward **Model Land Leasing Act–inspired reforms**.

EXAMPLES

- **Model Land Leasing policies** adopted in Andhra Pradesh and Uttarakhand.
- Digitized lease records enabled **credit access to sharecroppers**.
- Pilot leases included **climate clauses** for crop failure risk-sharing.

CONCLUSION

- India needs a **uniform, flexible, and legally secure** leasing framework to unlock land productivity.

Subtopic: Land Reforms

Card 1: Zamindari Abolition & Tenancy Reforms

QUOTE

"The land question is the heart of rural justice."

INTRODUCTION

- Land reforms like **zamindari abolition, tenancy regulation, and ceiling limits** were designed to democratize land ownership and "**redistribute dignity**" in rural India. Yet, implementation gaps persist in several states.

EXAMPLES

- Land ceiling redistribution remains below **2% of net sown area** nationally.
- Tenancy laws updated** in Odisha and UP to legalize sharecropping.
- Land rights regularization for **forest dwellers** implemented in tribal belts.

CONCLUSION

- Future reforms must balance **social equity with agricultural viability**, focusing on legal clarity and inclusiveness.

Card 2: Digital Land Records

QUOTE

"Jo zameen ka malik, wahi vikas ka adhikari."

INTRODUCTION

- Digitalizing land records through **SVAMITVA and DILRMP** enhances **ownership, dispute resolution, and rural credit flow**, aligning with the vision of "**smart villages with secure titles.**"

EXAMPLES

- Over **2 crore rural property cards** issued under SVAMITVA.
- States like Karnataka and Maharashtra fully digitized land records.
- GIS mapping** used to resolve multi-party claims in hilly regions.

CONCLUSION

- To sustain momentum, India must ensure **interoperable, tamper-proof, and citizen-friendly** land records systems.

Card 3: Agrarian Structure & Farmer Ownership

QUOTE

"True empowerment is tilling your own land with your own rights."

INTRODUCTION

- India's agrarian structure dominated by **small and marginal holdings** poses challenges to scale and investment. Empowering farmers through **land ownership and collective models** is key to rural transformation.

EXAMPLES

- Small and marginal farmers account for **86.2% of total operational landholdings**, but control only about **47% of the total crop area (10th Agriculture Census / 2024 DIU Estimates)**.
- SHG federations began **community farming on pooled land** in Kerala.
- Joint land titles under **PMAY-Gramin** improved women's land access.

CONCLUSION

- A shift toward **secure tenure, group farming**, and women's land rights can transform the rural landscape.

Card 4: LAND — Statistics & Land Leasing

INTRODUCTION

- India supports **16% of world population** and **20% livestock** on just **2.5% of geographical area**.
- Agriculture occupies the bulk of the land but its contribution to the national GVA has stabilized around **17-18% over the past few years**, highlighting ongoing structural shifts.
- Average land holding size: 1.08 ha.**
- Property disputes:** NITI Aayog — average **20 years** to settle a dispute.
- T. Haque Committee** — recommended **Model Land Leasing Act**.

DATA

- Avg landholding: **1.08 ha**.
- Property dispute resolution: **avg 20 yrs** — NITI Aayog.

Subtopic: Miscellaneous

Card 1: Urbanisation & Land Use Policy

QUOTE

"Land is not just for farming—it's the foundation of rural futures."

INTRODUCTION

- Urbanisation pressures are altering rural land use patterns. A **balanced, zoned, and inclusive** land use policy must anchor rural growth to the idea of **"planned prosperity."**

EXAMPLES

- National Geo-Spatial Policy** enabled rural land use planning tools.
- States like Gujarat adopted **village land use zoning** for agro-processing clusters.
- Land banks set up for **Urban clusters and MSME hubs**.

CONCLUSION

- India must ensure **smart land-use planning** that preserves agri-ecosystems while catalyzing rural industrialization.

SCIENCE & TECHNOLOGY

Section Group: Everyday Science & Innovations

Microtopic: Science and Technology- developments and their effects in everyday life

Subtopic: General Science

Card 1: S&T Applications

QUOTE

"Science is not only a tool for discovery—it is a lever for national development."

INTRODUCTION

- India is strategically applying science in sectors like **defence, healthcare, education, and energy** to empower governance and citizen services—building a **“Techade-ready Bharat”** that aligns knowledge with national priorities.

EXAMPLES

- i-Drone platform** deployed for vaccine delivery in remote areas.
- BrahMos-NG** and indigenous radar systems enhanced defence self-reliance.
- AI-based models integrated into **DIKSHA and NPTEL** platforms.

CONCLUSION

- Cross-sector innovation and **citizen-centric applications** are key to democratising scientific progress.

Card 2: GENERAL S&T — Overview

QUOTE

"The future depends on what we do in the present." — Mahatma Gandhi

"Jai Jawan, Jai Vigyan, Jai Anusandhan." — PM Modi

- Lab to field through **industry-academia collaboration** + incentivise adoption + facilitate **PPP in R&D**.
- **Regulatory sandbox** approach.

CONCLUSION

- **Balasya Mulam Vigyanam** (Science is the root of strength).
- Technology cannot be **uninvented** — must manage responsibly.

DATA

- **IMF (2024)** estimates almost **40% of global employment is exposed to AI**, rising to **60% in advanced economies**.

INTRODUCTION

- **RELEVANCE:** Technologies are especially meaningful for India at a time when ____ (IR 4.0).
- With solid **research base & workforce** founded on reliable govt support, India can harness **innovative applications** — stimulating economic growth & job creation.
- **RESPONSIBILITY:** Harness power of ____ responsibly, considering **ethical, safety and environmental implications**.
- **Issues:** Skill, Money, IPR, Security, Privacy, Divide, Ethics, Employment, Environment.

Card 3: Template 5: GENERAL S&T — Benefits of ____ Technology

FILL-IN-THE-BLANK INTRO

- "The benefits of ____ are especially meaningful for India at a time when _____. (IR 4.0)"

FILL-IN-THE-BLANK CONCLUSION

- "As we continue to harness the power of _____, we must do so responsibly, considering **ethical, safety and environmental implications**."

Subtopic: Innovative Technologies

Card 1: Role of Startups & R&D

QUOTE

"The next trillion-dollar ideas will come from India's labs and lanes."

INTRODUCTION

- With **STEM talent abundance** and policy boosters like **Startup India** and **Anusandhan NRF**, India is mobilising academia, private capital, and grassroots startups to transform into a **"global R&D powerhouse by 2047."**

EXAMPLES

- National Quantum Mission hubs** launched at IISER, IITs.
- Over **1000 DeepTech startups** supported via DST-BIRAC accelerators.
- Private labs contributed to **mRNA and AI diagnostics** in public health.

CONCLUSION

- Creating a **risk-tolerant, innovation-friendly ecosystem** is essential to sustain long-term R&D momentum.

Card 2: Emerging Frontier Technologies

QUOTE

"Emerging tech is the infrastructure of the invisible."

INTRODUCTION

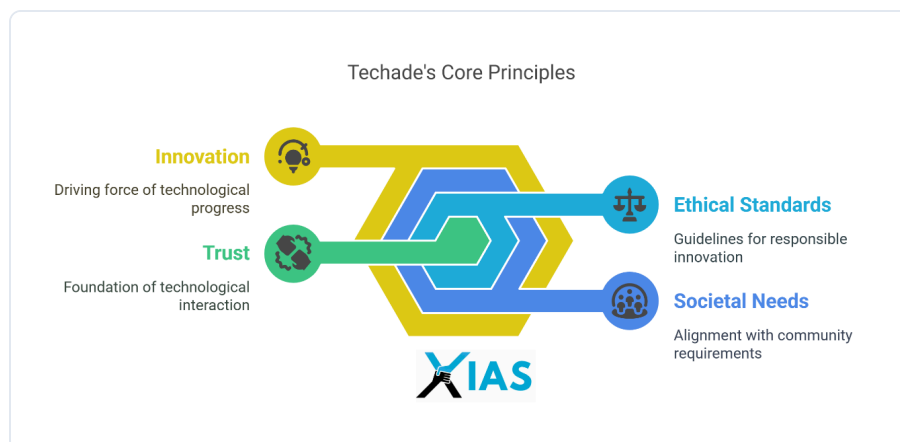
- India is leveraging **AI, IoT, Blockchain and Quantum** to catalyse next-gen governance and innovation—marking the shift to a **"Techade of Trust and Transformation"**, where data drives delivery and inclusion.

EXAMPLES

- IndiaAI Mission** launched with ₹10,000 crore outlay.
- Quantum Simulator** unveiled at IISc for research access.
- IoT sensors** used in Jal Jeevan Mission for real-time monitoring.

CONCLUSION

- Scalable pilots and **ethical frameworks** must anchor India's leap into emerging tech leadership.



Card 3: INDUSTRY REVOLUTION 4.0 (IR 4.0)

QUOTE

"In the fourth industrial revolution, it's not about the big eating the small, it's about the fast eating the slow."

INTRODUCTION

- **Definition:** Ongoing era of technological advancement characterised by the **fusion of digital, biological and physical technologies** — the "**disruptive dozen**".
- **Balasya Mulam Vigyanam.**

Card 4: DRONES

INTRODUCTION

- **Definition:** Aircraft without human pilot — **Unmanned Aerial Vehicles (UAVs)**.
- Recently **Drone (Amendment) Rules 2022** notified — abolished requirement of **drone pilot license**.
- **Steps:**
 - 1. **Digital Sky platform.**
 - 2. **PLI scheme.**
 - 3. **Mission Drone Shakti (DrAAS).**
 - 4. **Anti-Drone Tech by DRDO.**
 - 5. **Drone corridors.**
- **Applications:**
 - 1. **Healthcare (TL 'Medicine from Sky').**
 - 2. **KISAN Drones, SWAMITVA.**
 - 3. **Law & order.**
 - 4. **Delivery (Amazon Prime Air).**
 - 5. **DM (2022 Assam floods).**
 - 6. **Security (swarm drones, loitering munition).**
 - 7. **Law enforcement (illegal mining).**

Section Group: Frontier Technologies & IPR

Microtopic: Awareness in the fields of IT, Computers, and robotics

Subtopic: Artificial Intelligence

Card 1: Data Governance & IndiaAI Mission

QUOTE

"Data is the new oil—if governed, not spilled."

INTRODUCTION

- With policies like the **National Data Governance Framework** and **IndiaAI**, India aims to democratise datasets while ensuring privacy building a **"Digital Democracy with Responsible Intelligence."**

EXAMPLES

- Launch of **India Datasets Platform** for open innovation.
- **AI Centres of Excellence** approved in Tier-II cities.
- Data anonymisation protocols piloted for health analytics.

CONCLUSION

- Balancing **open access with secure boundaries** is key to harnessing India's digital capital.

Card 2: ARTIFICIAL INTELLIGENCE (AI)

INTRODUCTION

- **AI** = Simulation of human intelligence — capacity to **learn & make decisions** (pattern recognition, predictive analysis).
- **Machine Learning**: Subset using statistical learning algorithms.
- **Deep Learning**: ML technique inspired by human brain — relies on **Artificial Neural Networks**.
- **3 Cognitive Skills**: Learning, Reasoning, Self-correcting.
- **Applications**:
 - 1. Health (3Nethra eye scanner).
 - 2. Agri (autonomous farming, pest attack prediction, KN APMC price forecasting).
 - 3. Education (Robokart adaptive learning).
 - 4. Smart cities.
 - 5. Marketing (Alexa).
 - 6. SCM.
- **ASILOMAR Principles** — guiding light for AI development.
- **AI for ALL** principle.

CONCLUSION

- **AI is poised to add \$500 Bn to India's GDP by 2025** (10% of \$5 Trillion target). Regulation must balance **innovation vs. unintended consequences** (privacy, digital divide).

DATA

- AI GDP contribution: **\$500 Bn** projected by 2025.

Card 3: GENERATIVE AI

INTRODUCTION

- **Definition:** Type of AI producing **realistic content** (text, images, audio) based on training data.
- Powered by **GAN (Generative Adversarial Networks) + Transformers, LLMs, Deep Learning**.
- **Examples:** ChatGPT, DALL·E.
- **Applications:**
 - 1. ChatBots.
 - 2. DeepFakes.
 - 3. Dubbing.
 - 4. Product demonstration.
 - 5. Designing.
- **India Inequality Report (CMIE): IMAI-Kantar 2024 Report:** Urban internet penetration: **72%** vs Rural: **53%**.

CONCLUSION

- **"Tech is a double-edged sword."** — Alan Moore. Clerical jobs at greatest risk. Need **international high table discussion**.

DATA

- Urban internet penetration: **72%** vs Rural: **53%**.

Subtopic: IT

Card 1: IT in Governance

QUOTE

"When governance becomes digital, dignity becomes universal."

INTRODUCTION

- Through **Digital India, DBT**, and e-Gov platforms like **UMANG**, India is making governance **faceless, paperless and cashless**—ensuring **"minimum government, maximum governance"** becomes a lived reality.

EXAMPLES

- ₹30+ lakh crore transferred via DBT to over 90 crore beneficiaries.
- **DigiLocker** enabled 230+ crore document issuances.
- + services integrated under **UMANG app**.

CONCLUSION

- Interoperability and **citizen trust** must remain central as digital public infrastructure expands.

Card 2: SEMICONDUCTOR

INTRODUCTION

- **Definition:** Materials with conductivity between conductors & insulators — the **heart & brain of all modern electronics**.
- India **imports all chips** currently.
- **Steps:**
 - 1. **India Semicon Mission**.
 - 2. **PLI & DLI** schemes.
- **Issues:** Highly concentrated global supply chain (**47% with US**).
- **Suggestions:**
 - 1. BEL & HAL as fab foundries.
 - 2. Focus on **ATP** (Assembly, Testing & Packaging).
 - 3. Leverage **QUAD semiconductor supply chain initiative** for tech transfer.

CONCLUSION

- Steps like India Semicon Mission, PLI & DLI taken — **implementation will be key**.

DATA

- India's semiconductor market is projected to reach **\$109 Bn by 2030 (India Electronics and Semiconductor Association)**.

Card 3: CYBER PHYSICAL SYSTEMS (CPS)

INTRODUCTION

- **Definition:** Systems where **mechatronic systems are controlled or monitored by computer-based algorithms**.
- **Features:**
 - 1. Miniaturisation.
 - 2. Communication networks (devices talking).
 - 3. Computation ability.
 - 4. **IoT**.
- **Examples:**
 - 1. Autonomous cars (collision avoidance).
 - 2. Robotic surgery (precision).
 - 3. Smart grids.

CONCLUSION

- **National Mission on Interdisciplinary Cyber Physical Systems** will help augment human capabilities (prosthetics) & bring large-scale efficiency.

Card 4: 5G / 6G

INTRODUCTION

- **Definition:** 5th/6th generation wireless communication technology designed to **connect everyone & everything**.
- **Steps:**
 - 1. **National Digital Communication Technology Policy 2018.****
 - 2. ****NOFN.**
- **Applications:**
 - 1. Smart cities.
 - 2. EGov.
 - 3. Automated factories.
 - 4. Seamless coordination (NATGRID).
 - 5. Entertainment & multimedia (Immersive Multiverse).
- **Issues:**
 - 1. Low fibre backhaul (**80% towers on microwave backhaul**).
 - 2. IPR blocking tech access.
 - 3. Hardware (imported phones).
 - 4. Security concerns.

CONCLUSION

- Success depends on govt's ability to bridge **digital divide** & telecom companies' viability to invest.

DATA

- Cost of data: ₹300 → ₹10 per GB.

Card 5: QUANTUM COMPUTING

INTRODUCTION

- **Definition:** Application of collective quantum properties (**superimposition & entanglement**) to achieve objectives far beyond traditional capacity — uses **Qubits**.
- **Steps:**
 - 1. **Quantum research facility at Mhow.**
 - 2. **QUEST** (Quantum Enabled S&T Prog 2018).
- **Advantages (SSSP):** Speed, Storage, Simulation, Privacy.
- **Applications:**
 - 1. Research (Genome India).
 - 2. Secure Communication (**MICIUS — China**).
 - 3. Pharma.
 - 4. Meteorology.
 - 5. Data processing (consumer behaviour, financial markets).

DATA

- **Google Sycamore:** 200 seconds vs supercomputers' **10,000 years** — Quantum Supremacy.

Card 6: BLOCKCHAIN

INTRODUCTION

- **Definition:** Decentralised, immutable digital ledger distributed across a network — transactions anonymous & secure without 3rd party.
- **Steps:**
 - 1. IndiaChain by NITI Aayog.
 - 2. National Strategy on Blockchain 2021 (MeitY).
 - 3. Dubai experience.
- **Applications:**
 - 1. SCM (IBM Food Trust).
 - 2. Medical records (**ABDM**).
 - 3. Property records (**NRLMP**).
 - 4. Asset protection (**NFT**).
 - 5. Good governance.
 - 6. Anti-corruption (procurement, DBT).
 - 7. Voting.

Card 7: WEB 3.0 & METAVERSE

INTRODUCTION

- **Web 3.0** — New iteration of the World Wide Web.
- **Metaverse** — 360° immersive experiences integrating **physical, augmented & virtual reality** in a shared online space.

Card 8: CRYPTOCURRENCY

INTRODUCTION

- **Definition:** Digital/virtual money using **cryptography** for secure transactions, operating **independently of traditional financial institutions** (e.g., Bitcoin).
- **Features (ABCDEF):** Anonymous, Blockchain-based, Cross-border, Decentralised, Encrypted, Fast.
- **Suggestions:**
 - 1. Regulatory sandbox.
 - 2. Stablecoin.
 - 3. FSB international collaboration.

CONCLUSION

- Regulation must balance... **CBDC is in right direction.**

DATA

- **RBI:** Warns next global crisis may come from crypto.

Card 9: NFT (Non-Fungible Token)

INTRODUCTION

- **Definition:** **Unique digital assets** representing ownership of a specific item or content using **blockchain technology** to verify authenticity.

CONCLUSION

- Same as Cryptocurrency card.

Card 10: CBDC (Central Bank Digital Currency)

INTRODUCTION

- **Definition:** **Legal tender** issued by Central Bank in digital form using **blockchain platform**.
- **Applications:**
 - 1. Financial inclusion.
 - 2. Smart contracts.

CONCLUSION

- Need for wider **stakeholder participation** in rigorous analysis of pilot project outcomes by **RBI** to ensure full operationalisation in near future.

Card 11: IT (Information Technology)

INTRODUCTION

- **Definition:** Use of computers, software and networks to **handle and share information efficiently**.
- **IT + IT = IT** (Indian Talent + Information Technology = India Tomorrow).

CONCLUSION

- The vision of a **developed & digitally empowered India by 2047** calls for urgent investments in _____.

Card 12: Template 7: S&T — IT Conclusion

FILL-IN-THE-BLANK CONCLUSION

- "The vision of a **developed & digitally empowered India by 2047** calls for urgent investments in _____."

Microtopic: Awareness in the fields of Space and Defence technologies

Subtopic: Defence Technology

Card 1: DEFENCE INDIGENISATION

INTRODUCTION

- **Definition:** Process of developing and producing essential military capabilities **within the nation's own borders**.
- **SIPRI:** India remains the **world's largest arms importer (SIPRI 2024)**, but domestic defence exports reached a record **₹21,083 crore in FY2023-24**.
- **60% of weapons of foreign origin** — CAG.
- **Steps:**
 - 1. Defence Corridors (Brahmos in Lucknow).
 - 2. **AK-203 MIL**.
 - 3. **DPEPP 2020** (Defence Production & Export Promotion Policy).
 - 4. **DAC**.
 - 5. **FDI 74% automatic**.
- **Issues:**
 - 1. INS Vikrant delay (**2009–2021**).
 - 2. DRDO lacks specialisation.
 - 3. **Acquisition Trilemma** (time, cost, quality).
 - 4. Pvt sector — only **20% share** vs DPSUs **80%**.
 - 5. Capex spending only **25% of defence budget**.
- **Solutions:**
 - 1. **Dhirendra Singh Committee** — prioritise Buy (Indian IDDM).
 - 2. Long-term perspective plans.

CONCLUSION

- With a looming **two-front war threat**, ensuring **Atmanirbharta** in defence is an urgent necessity — balancing **urgency vs autonomy**.

DATA

- India remains the **world's largest arms importer (SIPRI 2024)**, but domestic defence exports reached a record **₹21,083 crore in FY2023-24**.

Card 2: SPACE DEFENCE

INTRODUCTION

- Recently launched **Mission DefSPACE** — private participation in space defence.
- USA has a dedicated **Space Force**.

CONCLUSION

- Build credible **defensive & offensive capabilities** + push for **comprehensive & binding international agreement** on peaceful use of Outer Space.

Subtopic: Space Technology

Card 1: SPACE SECTOR

INTRODUCTION

- Recently Govt released **Indian Space Policy 2023**.
- Vision of India as the "**launchpad of the world**".
- **95% of global space economy** is in **ground-based operations & space-based services** — neglected in India due to excessive focus on launch & manufacturing (**5%**).

DATA

- India's space economy: **\$8 Bn (approx 2% global share)** → projected to reach **\$44 Bn by 2033 (IN-SPACE 2023 Vision)**.

Card 2: PRIVATE PLAYERS IN SPACE

INTRODUCTION

- ISRO occupies a meager **2% share (\$5 Bn)** in the **global space economy of \$450 Bn** — indicating huge potential for private participation.
- Example: **Agnikul** — private space startup.
- **Draft Space Activities Bill 2017**.
- 1. **INSPACE** — interface between private players, ISRO & regulator.
- 2. **NSIL** (aggregator).
- **ISRO Budget**: only **\$1.5 Bn**.

CONCLUSION

- Indian space economy will become **\$60 Bn by 2030** — a golden opportunity.

DATA

- 1. Currently **30K satellites** in orbit.
- 2. Projected **80K by 2032**.
- ISRO: **\$50K/kg** vs SpaceX: **\$3K/kg**.
- Private sector access opened since **2000**.

Card 3: INDIAN SPACE POLICY 2023

INTRODUCTION

- Recently released — targeting **\$60 Bn space economy** from present **\$5 Bn**.
- **Pvt participation**: NSIL as aggregator.
- **Regulatory body**: InSPACE.
- **Nodal Dept**: **DOS** — policy guidelines.
- **Rationalisation of ISRO's role** — focus on R&D.
- End-to-end private activities allowed: ground stations, launch vehicles etc.

DATA

- Target: **\$60 Bn** space economy.

Card 4: SPACE DEBRIS

INTRODUCTION

- **Definition:** Defunct human-made objects and fragments that **orbit the Earth**.
- India's **NETRA** — for situational space awareness.
- **UK — RemoveDEBRIS mission.**
- **Indian ASAT test** — produced **400 pieces** of space debris.

DATA

- **100 million pieces** of debris >1mm in space.
- **2009 accident** — US & Russian satellite collision.

Card 5: ADITYA L1

INTRODUCTION

- **Space-based solar observatory** placed at **Lagrange Point L1**.
- **Payloads:**
 - 1. **SUIT** (Solar Ultraviolet Imaging Telescope).
 - 2. **ASPEX**.
- **Applications:**
 - 1. Nuclear energy.
 - 2. GPS.
 - 3. Satellite radio communication (solar storm monitoring).

CONCLUSION

- The mission is not just about **unraveling cosmic mysteries** but pivotal to India's **vision of space leadership**.

Card 6: CHANDRAYAAN-3

INTRODUCTION

- **Successful soft landing on the South Pole** — India became the **4th nation** (after US, Russia, China).
- **PM Modi**: "Win for humanity & science".
- **ARTEMIS mission** provides opportunity for collaboration.
- **Why South Pole?**:
 - 1. Unexplored regions.
 - 2. Permanently shadowed (possible water ice).
 - 3. Cryo-preserved evidence.
- **Payloads**:
 - 1. ChaSTE (thermal mapping).
 - 2. RAMBHA (plasma density).
 - 3. ILSA (lunar seismic activity).
- **Components**: Propulsion module + Vikram (lander) + Pragyan (rover).
- **Improvements from C2**:
 - 1. "Failure-based design" from "success-based design".
 - 2. Wider landing area.
 - 3. Strengthened legs.
 - 4. Increased fuel.
 - 5. Additional instruments.

DATA

- India — **4th nation** to achieve lunar soft landing.

Card 7: GAGANYAAN

INTRODUCTION

- ISRO confirmed **unmanned mission** — making India the **4th nation** to send humans to space.
- **Features**:
 - 1. Circle Earth for **5–7 days at LEO (300–400 km)**.
 - 2. **3 flights** total (**2 unmanned + 1 human** with 3 astronauts).
 - 3. Crew + habitable orbital module + life support + **Human-rated GSLV Mk 3** + re-entry, recovery & escape/abort system.

CONCLUSION

- (No specific conclusion provided — same template as Aditya L1)

DATA

- Orbit: **300–400 km LEO**.

Card 8: Template 6: ADITYA L1 / GAGANYAAN — Mission Conclusion

FILL-IN-THE-BLANK CONCLUSION

- "The ___ mission is not just about **unraveling cosmic mysteries** but also **pivotal to India's vision of space leadership**."

Microtopic: Awareness in the fields of nano-technology, bio-technology and issues relating to intellectual property rights

Subtopic: Bio-Technology

Card 1: BIOTECHNOLOGY

INTRODUCTION

- **Definition:** Use of **biological systems & living organisms** to develop various products.
- **India:**
 - 1. Administered **220 crore vaccines**.
 - 2. Eliminated smallpox, polio.
- **Best Practice:** **COVID Suraksha Program** — World's **first COVID nasal vaccine (BB)** — success of state-led R&D.
- **Steps:**
 - 1. **BIRAC**.^{**}
 - 2. **National Biotech Development Strategy 2015–2020**.
 - 3. ****National Biopharma Mission**.
- **Applications:**
 - 1. Agri (GM, sex-sorted semen).
 - 2. Nutrition (bio-fortification).
 - 3. Environment (bioremediation, Biodigestor).
 - 4. Industry (biodegradable plastics, biofuels, enzymes).
 - 5. Health (vaccines, ELISA, stem cell therapy, **recombinant DNA tech**).
- **Issues:**
 - 1. Biological weapons/bioterrorism.
 - 2. Biopiracy.
 - 3. Biosafety.
 - 4. Patents.

CONCLUSION

- Despite growth, India's share is only **3% of global biotech industry** — huge potential.

DATA

- India administered **220 crore vaccines**.
- India's bioeconomy reached **\$150 Bn in 2023**, targeting **\$300 Bn by 2030**.

Card 2: GM CROPS

INTRODUCTION

- Recently Govt approved **environmental release of GM Mustard Hybrid DMH-11**.
- **Definition:** Changing characteristics of an organism by **transferring DNA from another organism**.
- India has the **5th largest area** under GM cultivation.
- **Precautionary Principle** must guide all GM decisions.
- Alternative: "**Safe agri-ecological solutions**" — e.g., System of Mustard Intensification.

DATA

- **DMH-11: 20–25% higher yield** (current yield is ½ of global average).
- **BT Cotton:** Reduced insecticide use by **>50%** in AP & GJ.
- 1. Only **29/170 countries** grow GM crops.
- 2. **90% in just 5 countries**.

Subtopic: Intellectual Property Rights

Card 1: IPR & Deep Tech Regulation

QUOTE

"Innovation without regulation is chaos; regulation without trust is paralysis."

INTRODUCTION

- India is refining its **IPR, localisation, and deep tech frameworks** to protect creators and ensure safe innovation crafting a "**sandbox governance model**" fit for AI, blockchain and future tech frontiers.

EXAMPLES

- India ranks **38th among the 139 economies featured in the Global Innovation Index (GII) 2025**.
- **Data localisation norms** enforced for fintech and health apps.
- DeepTech Task Force formed under DPIIT for policy roadmap.

CONCLUSION

- Agile regulations, **global alignment and innovation autonomy** must co-exist to future-proof India's digital rise.

Card 2: IPR (Intellectual Property Rights)

INTRODUCTION

- **Definition:** Legal protections for **original creations, inventions and ideas**.

CONCLUSION

- Reform IPR ecosystem to balance **incentives for cutting-edge research** with **larger public good**.

Subtopic: Nano Technology

Card 1: NANOTECHNOLOGY

INTRODUCTION

- **Definition:**
 - 1. Field of research at scale of **atoms & molecules** at **nanometer range (1–100 nm)**.
 - 2. 1 nm = 1 billionth of a metre.
- **Steps:**
 - 1. **Nano S&T Initiative (NSTI).****
 - 2. ***Nano Mission 2007.***
 - 3. ****Visheswaraya PhD Scholarships.**
- **Applications:**
 - 1. Agri (nanosensors, **nano urea**, precision farming, nanogels).
 - 2. Health (controlled drug delivery, remote monitoring, **nanobots**).
 - 3. Space.
 - 4. Energy (solar paints, nanogenerators).
 - 5. Construction.
 - 6. Electronics (nano batteries).
 - 7. **Defence (stealth).**

CONCLUSION

- **Lab to field** approach needed.

Section Group: Indigenous Tech & Achievements

Microtopic: Achievements of Indians in S&T; Indigenisation of technology & developing new

technology

Subtopic: Achievements of Indians in S&T

Card 1: Indigenous Innovations (DRDO, ISRO)

QUOTE

"Aatmanirbharta in science is the cornerstone of strategic autonomy."

INTRODUCTION

- Institutions like **DRDO, ISRO, and CSIR** are steering India toward **indigenous R&D ecosystems**, with innovations that serve both security and societal missions—reflecting a shift to **“lab-to-field” excellence for self-reliant India**.

EXAMPLES

- DRDO unveiled **Agni-5 MIRV missile** for credible deterrence.
- ISRO launched **INSAT-3DS** for advanced meteorological data.
- CSIR developed **bio-based nanofertilizers** for sustainable farming.

CONCLUSION

- Strengthening institutional synergy and **tech-commercial linkages** will boost India’s global scientific standing.

Card 2: Indigenisation of Technology

QUOTE

"Innovation isn't just imitation with delay—it is ownership with intent."

INTRODUCTION

- From semiconductors to defence platforms, India is accelerating **tech indigenisation** through Make in India, Strategic Technology Boards, and **FDI-linked innovation hubs**—aligning with the vision of **“design in India, for the world.”**

EXAMPLES

- India approved **semiconductor fabs** with \$10B outlay under Digital India push.
- Indigenous LoRa-based chips deployed in **IoT agriculture pilots**.
- **Tejas Mk-1A fighter jets** cleared for production by HAL.

CONCLUSION

- Robust IP ecosystems and **domestic capacity-building** are vital for genuine tech sovereignty.

Card 3: Government R&D Policies

QUOTE

"Good science needs great policy to thrive."

INTRODUCTION

- India's evolving **STI Policy**, **National R&D Policy**, and budgetary prioritisation reflect a **commitment to science-led governance**, with a push toward **"Jan Bhagidari in Innovation"** and equity in research opportunities.

EXAMPLES

- STIP 2020 draft** promotes inclusivity and open-access platforms.
- Allocation for **R&D in clean energy** doubled under Union Budget.
- Innovation Clusters** launched under State S&T councils.

CONCLUSION

- Forward-looking R&D policy must blend **public accountability and private agility** to deliver real-world impact.

Card 4: CRYOGENICS

INTRODUCTION

- Definition:** Production and behaviour of materials at **very low temperatures** (below **−153°C**).
- Applications:**
 - 1. MRI.
 - 2. Freeze drying.
 - 3. Cryosurgery.
 - 4. Cold treatment for tools (metallurgy).
 - 5. **Cryo engines** (liquid H₂, O₂).

CONCLUSION

- (No specific conclusion provided — refer general S&T template)

DATA

- Temperature threshold: below **−153°C**.

Card 5: WOMEN IN STEM

QUOTE

"Our thinking is not merely to empower women through science but also empower science by the contribution of women."

INTRODUCTION

- Ritu Karidhal** — "Rocket Woman of India" — key to **Chandrayaan** mission.

Card 6: Template 8: S&T — Scientific Personality Intro

FILL-IN-THE-BLANK INTRO

- "____ was a great scientific luminary, whose profound contributions in the field of ____ have reshaped our understanding and spurred unprecedented advancements."

FILL-IN-THE-BLANK CONCLUSION

- "____'s contributions have left an **indelible mark** in the field of ____ and continue to influence countless advancements & innovations that shape our modern world."

Card 7: Template 10: CRYOGENICS — General S&T Template Reference

FILL-IN-THE-BLANK

- Refer General S&T Template (TEMPLATE 5) for intro/conclusion.

APPLICATIONS TO FILL IN

- 1. MRI.
- 2. Freeze drying.
- 3. Cryosurgery.
- 4. Cold treatment for metallurgy.
- 5. **Cryo engines (liquid H₂, O₂).**

ENVIRONMENT

Section Group: Climate Change

Microtopic: Global warming, climate change, and international frameworks

Subtopic: Climate Change

Card 1: CRYOSPHERE

INTRODUCTION

- **Definition:** Frozen components of the Earth system at or below land and ocean surface — e.g., **permafrost**.
- **Permafrost holds 2× as much carbon** as the atmosphere.
- Key glaciers at risk: **Thwaites (Antarctica), Larsen C, Okjökull (Iceland)**.

DATA

- **10% of world population** lives in high mountains.
- **IPCC: 80% of glaciers** expected to be lost by **2100**.

Card 2: OCEANS — Acidification & AMOC

INTRODUCTION

- **Ocean Acidification:** Decrease in ocean pH below **8** due to **CO₂ uptake**.
- **AMOC** (Atlantic Meridional Overturning Circulation) — the Atlantic branch of the global **ocean conveyor belt**, distributing heat and nutrients.

DATA

- **IPCC:** AMOC is "**very likely to decline**" over the 21st century.

Card 3: CLIMATE CHANGE — Overview & India's Commitments

QUOTE

"If solutions within the system are so impossible to find, we should change the system itself." — Greta Thunberg

"IPCC reports are code red for humanity." — Antonio Guterres

BEST PRACTICE — MEENANGADI VILLAGE (KL)

- 1. Carbon-neutral panchayat.
- 2. GHG inventory.
- 3. **Tree Bank Project** — loans as per trees planted (no collateral).
- 4. Panchayat pays interest.
- **Adaptation — NEED: North European Enclosure Dam.**

DATA

- **Atlas of Energy:**
 - 1. China — **10 Gt CO₂**.
 - 2. USA — **5 Gt**.
 - 3. India — **2.3 Gt**.
- **Per capita:**
 - 1. US — **14 MT**.
 - 2. China — **7 MT**.
 - 3. India — **1.78 MT**.
- India has secured the **10th rank in the Climate Change Performance Index (CCPI) for 2025** (Germanwatch).
- 1. India **7th most vulnerable** country — Climate Risk Index (CRI).
- 2. **#1 in fatalities, #2 in monetary losses.**
- **1 person in USA = 108 people in Ethiopia** (carbon emissions).
- **Carbon-negative countries:** Bhutan.
- **IPCC:** To achieve **1.5°C target** — **43% reduction in GHG by 2030** (relative to 2019).
- **MoES Assessment:**
 - 1. Heatwaves to become **3–4× by 2100**.
 - 2. Drought frequency to **double in next decade**.
 - 3. Sea level to rise **3.3 mm/year**.
- Healthcare cost of climate change: **8.5% of GDP**.
- **ILO:** Productivity loss — **34 million full-time jobs**.
- Sea level rise — **20% of global assets** exposed.

INTRODUCTION

- **Definition:** Significant changes in global temperature, precipitation, wind patterns over **several decades**.
- **Panchamrit (India's 5 commitments):**
 - 1. Net zero by **2070**.
 - 2. **Non-fossil energy — 500 GW by 2030**.
 - 3. **50% renewable energy by 2030**.
 - 4. Reduce total carbon emission by **1 bn tonnes by 2030**.
 - 5. Reduce carbon intensity by **45% by 2030**.
- **Climate Paradox:** China — highest emitter but part of G77.
- **Energy Trilemma:** Energy equity + Energy security + Environmental sustainability.

- **Key Keywords:**
 - 1. 6th mass extinction.
 - 2. DALY.
 - 3. Degrowth.
 - 4. LiFE.
 - 5. Circular economy.
 - 6. Carbon colonialism.
 - 7. Climate denialism.
 - 8. Greenwashing.
 - 9. Hothouse.
 - 10. Tipping point.

Card 4: NET ZERO

INTRODUCTION

- **Definition:** Balance where GHG emissions are **offset by equal absorption** from the atmosphere.
- **India's target:** Net zero by **2070**.
- **New NDCs:**
 - 1. Reduce emission intensity by **at least 45%**.
 - 2. **50% non-fossil fuel energy**.
 - 3. **2.5–3 bn tonnes** of additional carbon sink.
- **Steps:**
 - 1. **OSOWOG**.
 - 2. **IRIS** (Infrastructure for Resilient Island States).**
 - 3. ****CDRI**.

DATA

- India net zero target: **2070**.

Card 5: LT-LEDS (Long-Term Low Emission Development Strategy)

INTRODUCTION

- Lays out steps to achieve **net zero by 2070**:
- 1. **Low carbon electricity:** 50% non-fossil; RPOs; **3× rise in nuclear capacity by 2032**.
- 2. **Transport:** 20% ethanol blending; **railways net zero by 2030**.
- 3. **Urban efficiency:** National Urban Policy Framework.
- 4. **Innovation + low emission industry:** Green H₂; National Mission for Enhanced Energy Efficiency.
- 5. **CO₂ removal:** CCUS.
- 6. **Forest cover:** NDC target — **2.5–3 bn tonnes CO₂ equivalent by 2030**; Green India Mission.
- 7. **Economic & financial:** business low-carbon strategies.

Card 6: AGRICULTURE & CLIMATE CHANGE

INTRODUCTION

- Agriculture contributes **18% of GHG emissions**.
- **70% of agricultural energy usage** is in irrigation.
- **ES 2018**: CC will cause farmer income loss of **15–18%** in irrigated areas and **20–25%** in unirrigated areas.
- **GM Crops context**:
 - 1. Only **29/170 countries** grow GM crops.
 - 2. **90% in 5 countries** (USA, Canada, Brazil, Argentina, India).
 - 3. India has **5th largest area** under GM cultivation — **11 mn ha**.

DATA

- Agri contributes **18% of GHG**.
- Income loss projected: **15–25%** due to CC.

Card 7: CARBON TRADING MARKET

INTRODUCTION

- **Definition**: Market-based approach to reducing GHG emissions — provides **economic incentives** for industries to limit emissions.
- **Issues**:
 - 1. Double counting.
 - 2. Low market transparency.
 - 3. **Greenwashing**.
- **India**:
 - 1. **REC (Renewable Energy Certificates)**.**
 - 2. **Energy Saving Certificates under PAT** (Perform Achieve Trade).**

CONCLUSION

- Gradually move from **PAT to Cap and Trade** system.

Subtopic: IPCC Reports

Card 1: IPCC REPORT — Key Findings

INTRODUCTION

- Avg annual emissions **2010–19 higher than any previous decade** — but rate of growth slowed.
- Global share of emissions from **urban areas is increasing**.
- **Suggestions:**
 1. Mitigation efforts to be embedded within **wider development context**.
 2. **Equitable partnerships** — civil society, political actors, businesses, youth.
 3. Avoid **Maladaptation** — inclusive planning + biodiversity + autonomous adaptation.
 4. **Climate resilient development**.
- **Impact on food production:**
 1. 69% Indian marine fisheries impacted by CC.
 2. Rice production can decrease by **10–30%**.
 3. Maize by **25–70%**.

CONCLUSION

- **"A brief and rapidly closing window of opportunity to secure a liveable and sustainable future for all."**

DATA

- **2019 — India: 4 mn disaster-induced displacements.**
- Rice:
 1. **10–30%** decline.
 2. Maize: **25–70%** decline.

Subtopic: UNFCCC

Card 1: Climate Agreements & UNFCCC

QUOTE

"Climate action is not charity—it is enlightened self-interest."

INTRODUCTION

- India's **Panchamrit pledges**, participation in **COPs**, and the launch of the **LiFE Mission** signal a pivot to "**pro-planet lifestyles**" and equitable climate diplomacy rooted in **climate justice for the Global South**.

EXAMPLES

- India co-led the **International Solar Alliance** and **CEM missions**.
- LiFE campaigns scaled in 1 lakh+ schools and colleges.
- INDCs updated to include **45% emissions intensity cut**.

CONCLUSION

- Low-carbon growth must anchor India's **development-climate alignment** in the coming decades.



Card 2: CLIMATE JUSTICE

INTRODUCTION

- **Pakistan floods** — cost **>\$30 billion, 1,700 casualties**.
- **58 climate-vulnerable countries** responsible for only **5% of global emissions**.

DATA

- Pakistan floods cost: **>\$30 billion, 1,700 casualties**.
- **58 vulnerable countries** → **5% of emissions**.

Card 3: COP 27 — Sharm El-Sheikh

INTRODUCTION

- **Sharm El-Sheikh Implementation Plan:**
 1. **Breakthrough in L&D** (Loss & Damage).
 2. Proposed **EWS to all population**.
 3. **FAST** — Food and Agriculture for Sustainable Transformation.
 4. **Koronivia Joint Work for Agriculture**.
 5. **ENACT** — Enhancing Nature-based Solutions for Accelerated Climate Transformation.
 6. **Mangrove Breakthrough** — protect **15 mn ha by 2030**.
 7. Called the "**African COP**".
- **Contentions:**
 1. No commitments to phase out all fossil fuels.
 2. Inadequate climate finance.
 3. Lack of clarity in L&D.
 4. Greenwashing.
 5. Fossil fuel lobby.
- **Indian Concerns:**
 1. Gap in equity/climate justice.
 2. Shrinking global carbon budget.
 3. Developmental needs of developing countries.
 4. **CBAM (discriminatory)**.
 5. Agriculture mitigation scope.
 6. Shortage in climate finance.

Card 4: CLIMATE FINANCE

INTRODUCTION

- **Definition:** Local, national or transnational funding from public, private and alternative sources for **climate change mitigation and adaptation**.
- **Keywords:**
 - 1. Carbon leakage.
 - 2. Coal cess.
 - 3. **CBAM by EU**.
 - 4. CBDR.
 - 5. Zero-sum game.
- **Efforts:**
 - 1. **GEF**.
 - 2. **GCF**.
 - 3. **Adaptation Fund**.
 - 4. **Catastrophe bonds**.
- **RBI Dy. Governor suggestions:**
 - Setup **GIIT** — Green Infrastructure Investment Trust (on lines of REIT/InvIT).
 - Taxonomy on **green finance** to avoid greenwashing.
 - Expertise to evaluate **climate risks** in green loan portfolios.

DATA

- **\$70–100 bn/yr** needed by developing countries for adaptation (2010–2050).
- **Copenhagen Accord: \$100 bn p.a. by 2020** — **<2% of promised funds given** in last 10 years.
- India needs **\$10 trillion** investment to achieve net zero by 2070.
- Opportunities:
 - 1. **\$1 investment = \$4 return**.
 - 2. **65 mn new green jobs** can be created.
- **3/4th** of adaptation projects have no direct co-relation.

Section Group: Conservation & Ecosystems

Microtopic: Conservation and environmental impact assessment

Subtopic: Biodiversity

Card 1: Biodiversity Conservation

QUOTE

"We do not inherit the Earth from our ancestors; we borrow it from our children."

INTRODUCTION

- India's approach to conservation through **Wildlife Protection (Amendment) Act, eco-sensitive zones, and corridor connectivity** reflects a shift to "**coexistence over conflict**"—a foundational ethic in its ecological governance.

EXAMPLES

- 98+ Eco-Sensitive Zones** notified around protected areas.
- Project Cheetah** rewilded species after 70+ years of extinction.
- Kaziranga landscape project** links riverine and forest corridors.

CONCLUSION

- Conservation must blend **community stewardship and scientific planning** for lasting biodiversity gains.

Card 2: Forest Cover & CAMPA Funds

QUOTE

"Forests are not just carbon sinks—they are cultural lifelines."

INTRODUCTION

- India's forest conservation efforts via **State of Forest Reports, CAMPA Funds**, and afforestation schemes—reflect a "**green GDP**" approach to balancing development and natural wealth.

EXAMPLES

- 24.6% of India's area** under forest as per latest report.
- ₹60,000+ crore allocated under **CAMPA** for compensatory afforestation.
- Urban forests developed under **Nagar Van Yojana**.

CONCLUSION

- Ecological restoration must focus on **quality of green cover and local livelihoods**, not just numbers.

Card 3: BIODIVERSITY

QUOTE

"The only thing that displaced more people than war is wildlife conservation." — Mark Dowie

CONCLUSION

- **Biodiversity is the reason for survival of humankind** — the world must collaborate to meet **Global Biodiversity Framework** targets.

DATA

- **IUCN: 1/3rd of species** threatened with extinction.
- **Living Planet Report: 70% wildlife population declined** since 1970.

INTRODUCTION

- 1. India is a **megadiverse country** — **2.4% of land** supports **7–8% of all recorded species**.
- 2. **4 of 36 biodiversity hotspots**.
- **Triple Planetary Crisis: CC + Pollution (DALY) + Biodiversity loss (6th mass extinction)**.
- **Amur Falcon Model** — "Predators to Protectors" — awareness generation to reduce poaching.
- **Millet Sisters Network** — protects indigenous crop diversity.
- **Evil Quartet:**
 - 1. **Habitat loss & fragmentation** — CC, desertification, developmental projects.
 - 2. **Over-exploitation** — extinction (Steller's sea cow); illegal trade.
 - 3. **Invasive species** — e.g., lantana; **10 invasive species in India — damage > ₹8 trillion**.
 - 4. **Co-extinction**.
- **ESZ (Eco-Sensitive Zones)** — ecologically fragile areas around protected areas under **EPA 1986**.

Card 4: CORAL REEFS

INTRODUCTION

- **Definition:** Large underwater structures of **calcium carbonate** made from coral skeletons — the **"tropical rainforests of the oceans"**.
- **Map:** Malvan, Mannar, Kutch, A&N, Lakshadweep.
- Key bodies:
 - 1. **ICMAM** (Integrated Coastal and Marine Area Management).**
 - 2. ***INCOIS** — *Coral Bleaching Alert System*.*
 - 3. ****BioRock**.
 - 4. **Reefwatch**.
- **Best Practice: Cryomesh** technology for coral restoration.

DATA

- 1. India home to **30% of corals**.
- 2. **50 crore people** depend on corals for livelihood and cultural identity.
- **6th Status of the Corals of the World Report: 14% of coral reefs lost since 2010**.

Card 5: WESTERN GHATS

INTRODUCTION

- **UNESCO World Heritage Site** spanning India's western coast — rich biodiversity and unique ecosystems.
- **Gadgil Committee**: recommended **100% as ESA** (Ecologically Sensitive Area).
- **Kasturirangan Panel**: recommended **37% as ESA**.

Subtopic: EIA

Card 1: EIA & Eco-Sensitive Zones

QUOTE

"Environment regulation is not anti-growth—it is pro-survival."

INTRODUCTION

- India's **EIA notifications**, **CRZ rules**, and **eco-zonal buffers** reflect a calibrated approach to development—aiming for a **"green clearance, not green compromise"** philosophy in project governance.

EXAMPLES

- Revised **CRZ norms** enabled coastal tourism with ecological safeguards.
- Online **PARIVESH portal** ensured faster EIA clearances with scrutiny.
- Over **200+ projects** modified based on ESZ impact assessments.

CONCLUSION

- Transparent, science-led EIAs** must become the norm, not the exception.



Card 2: ENVIRONMENTAL LAWS — EIA

INTRODUCTION

- **Definition:**
 - 1. Process/study that **predicts the impact** of a proposed project on the environment.
 - 2. Prevents approval without considering impact.
 - 3. Compares alternatives to minimise harm.
- **Flow:** Screening → Scoping → Impact Analysis → Mitigation → Draft EIA Report → **Public Hearing** → Revision & Approval → Implementation → Monitoring.
- **Negative Examples:**
 - 1. LG Polymers — no EIA for decades.
 - 2. Vadvan Port EIA — only **1 season's data** considered.
- **Kasturirangan Committee:**
 - 1. GRM for violation compliance.
 - 2. Capacity building.
 - 3. No activity in ESZ.
- **Way Forward: SEA (Strategic Environment Assessment).**

DATA

- **41% of NGT cases** related to **faulty EIA**.

Subtopic: Environment

Card 1: ENVIRONMENT — Importance

INTRODUCTION

- **Highly productive** — Supporting **Biodiversity**, shelter to various species.
- Acts as a **carbon sink**.
- **Regulating services:** Purification, micro climate, soil/coastal erosion control, **biogeochemical cycles & nutrient recycling**, disease regulation.
- **Provisioning services:** Fuelwood, local economy — **Fisheries, eco-tourism**.
- **Cultural services:** aesthetic, recreational, spiritual, tribal identity.
- Acts as **natural barriers** against disasters.
- Functions as a **Bioindicator** of ecosystem health.

Card 2: ENVIRONMENT — Challenges (Threats)

INTRODUCTION

- **Natural Threats:** Cyclone, Forest fire, Floods, **El Niño** / **Land use change**.
- **Manmade Threats:**
 1. **Deforestation** — Land use change / encroachment.
 2. **Habitat destruction** from **MATTIE** — Mining, Agri, Transport, Tourism, Industry, HEP.
 3. **Unsustainable tourism**.
 4. **Waste dumping / pollution** — agri run-off; industry; sewage; municipal solid waste; plastic.
 5. **Unsustainable urbanisation** — population explosion, consumerism, concretisation, linear economy.
 6. **Tragedy of commons / over-harvesting**.
 7. **Climate change**.
 8. **Invasive species**.

Card 3: ENVIRONMENT — Impact (All Spheres)

INTRODUCTION

- **Hydrosphere impacts:**
 - Surface water: **eutrophication, dead zones, contamination**.
 - Groundwater: **depletion**.
 - Open oceans: **sea level rise, acidification, de-oxygenation, temp rise, stratification, pollution, increased cyclones, AMOC disruption**.
 - Coastal: **receding/emerging coasts, saltwater intrusion**.
- **Atmosphere impacts:**
 - Extreme weather events, **global warming**, heat islands, **acid rain**, smog, **air pollution / AQI**.
- **Biosphere impacts:**
 - **6th mass extinction**, man-animal conflict, changing migration patterns, pest attacks, **bio-magnification & bio-accumulation**, co-extinction.
- **Lithosphere impacts:**
 - **Desertification**, erosion, salinity, soil moisture depletion.
- **Anthroposphere impacts:**
 - Food security, loss of livelihood, disaster (GLOF), pandemics/diseases, permafrost, refugee crisis, **social — PHEE WASH**, agri distress, loss of productivity, human health, water security.

Card 4: ECOSYSTEM — General Remedies

INTRODUCTION

- **Remedy for all ecosystems:**
- Policy:
 - 1. **Paris Agreement.**
 - 2. **Landscape approach.**
 - 3. **Secure Himalayas.**
- Human-centred: **ESZ, eco-tourism, community participation, Kasturirangan Committee, sustainable agriculture.**
- Technology: **drones, satellite monitoring, EWS.**

Card 5: FORESTS

QUOTE

"We have a choice — collective action or collective suicide." — Antonio Guterres

DATA

- **45%** of Earth's original forest lost in the last century.
- **UNFCCC: 80% deforestation** in tropical countries due to agriculture.
- FSI forest cover: **21.7%** vs NFP target of **33%**.

INTRODUCTION

- **FSI** — current forest cover: **21.7%** (target under **NFP 1988**: 33%).
- **3 lakh ha** of forest land diverted in 15 years — maximum for **mining, roads, irrigation.**
- **Best Practices:**
- **Eco-feminism:** Piplanatri model.
- **JFM (Joint Forest Management)** — Maldharis.

Card 6: ANTARCTICA

INTRODUCTION

- **Antarctica Treaty 1959:**
 - 1. No military activity, no nukes, no territorial claims.
 - 2. **Scientific research allowed.**
- **Indian Antarctica Bill 2021** — India's domestic legal framework.
- Indian research stations: **Dakshin Gangotri, Maitri, Bharati.**

DATA

- Antarctica holds **90% of world's ice** and **70% of world's freshwater.**

Subtopic: Environmental Laws

Card 1: ENVIRONMENTAL LAWS — FRA (Forest Rights Act)

INTRODUCTION

- **2023 Amendment:**
 1. Defines forest in an **objective manner**.
 2. **No prior approval** required for **strategic/security/defence projects** within **100 km of international border**.
 1. The act in its current form creates **hurdles in the way of development**.
 2. Yet any change must recognise the **symbiotic relationship between forest, biodiversity and stakeholders**.

CONCLUSION

- Change will only work if it recognises the symbiosis between **forest, biodiversity and stakeholders**.

Card 2: ENVIRONMENTAL LAWS — NGT (National Green Tribunal)

INTRODUCTION

- **PoNJ** (Principle of Natural Justice) as foundation.
- 1. **Ennore Oil Spill** — affected **>1 lakh fishermen**.**
- 2. **NGT ordered shipping company to pay ₹240 crore compensation**.**
- Criticisms:
 - 1. **Green hurdle**.
 - 2. **Selective adjudication**.

Card 3: ENVIRONMENTAL LAWS — PPV&FR Act

INTRODUCTION

- Key case: **PepsiCo case** — landmark instance of farmers' rights vs. corporate IP.

Card 4: ENVIRONMENTAL LAWS — WPA (Wildlife Protection Act)

INTRODUCTION

- India is the **cradle of wildlife** from the times of IVC — wildlife worshipped as gods and goddesses.
- **WPA 1972** enacted to **preserve and promote wildlife** of India.
- Proposed amendments are under consideration.

Card 5: ENVIRONMENTAL LAWS — BDA (Biological Diversity Act)

INTRODUCTION

- Enacted to ensure **conservation, benefit sharing and sustainable use** of biodiversity in India in line with **CBD** and protocols — **Cartagena (biosafety) & Nagoya (ABS)**.
- **Issues:**
 1. Commerce over conservation.
 2. Too many exceptions.
 3. **ABS** — Divya Pharmacy case controversy.

CONCLUSION

- **Biodiversity is the reason for survival of humankind** — the world must collaborate to make targets of the **Global Biodiversity Framework** into reality.

Subtopic: Sustainable Development

Card 1: Role of Tribunals (NGT, FRA)

QUOTE

"Justice delayed in ecology is extinction in slow motion."

INTRODUCTION

- The **National Green Tribunal** and **Forest Rights Act** have emerged as critical institutions in balancing ecology with equity grounding India's vision of **"Green Governance with Tribal Inclusion."**

EXAMPLES

- NGT stayed illegal mining near **critical elephant corridors**.
- FRA recognised **1.8 million tribal land titles** with livelihood security.
- NGT fined several states for **solid waste non-compliance**.

CONCLUSION

- Strengthening institutional capacity ensures **environmental justice reaches the last forest-dweller**.

Card 2: ENVIRONMENT — Solutions

INTRODUCTION

- **One Health** approach.
- **Practices (Bottom-up):**
 1. **Govt** — ESIA.
 2. Art. 48A.
 3. Art. 21.
- 1. **Individual** — **5R** (Refuse, Reduce, Reuse, Recycle, Repair).
- 2. **LiFE**.
- 3. Art. 51A.
- 1. **Business** — SDG 12.**
- 2. **Triple Bottom Line (People, Profit, Planet).**
- 3. **Pay for Ecosystem Services**.
- 4. ****ESG**.
- **Key Principles:**
 - Policy: **Wise use, Precautionary, Ecocentric**.
 - Justice: **CBDR-RC, L&D, Absolute Liability, Polluter Pays**.
 - Corporate: Triple Bottom Line, ESIA, **Access & Benefit Sharing** (fair earth share).
- **LT Low Carbon Development Strategy (COP 27)** — multi-sectoral measures for **net zero & climate resilience**.

CONCLUSION

- **Collective action vs collective suicide** — Antonio Guterres.
- **Saptrishi** — Green Growth.
- **Think Global, Act Local**.
- **Lab to Street**.

Card 3: SUSTAINABLE DEVELOPMENT

QUOTE

"The greatest threat to our planet is the belief that someone else will save it." — Robert Swan

"We won't have a society if we destroy the environment." — Margaret Mead

CONCLUSION

- ____ is not just an option but a **necessity for a sustainable future**.
- **Collective action vs collective suicide** — Antonio Guterres.

DATA

- India ranks **99th out of 167 countries on the 2025 SDG Index** (SDG index).
- India ranked **176th among 180 countries in the 2024 Environment Performance Index**.

INTRODUCTION

- **Sustainable development** = harmonious relationship between **development, social equity and environment**.
- India voted at **UNGA** to recognise **right to clean, healthy and sustainable environment** as a human right.
- **Triple Emergency**: CC + biodiversity loss + pollution.

Card 4: MISSION LiFE

QUOTE

"LiFE does not look at which country is in coalition but brings everyone together."

"Mission LiFE is an important step towards renewed and reimagined multilateralism."

INTRODUCTION

- **Objective:** Promote environmentally conscious lifestyle — "**mindful & deliberate**" utilisation instead of "wasteful & mindless consumption".
- India-led **global mass movement** to nudge individual and collective action.
- **>80%** of all **RLB and ULB** to become environment-friendly by **2028**.
- **Change in demand** → **Change in supply** → **Change in policy**.
- **Best Practice:** Solar-powered village of **Modhera**.

DATA

- **>80% RLB and ULB** to be environment-friendly by **2028**.

Card 5: Template 1: COMMON INTRO/CONCLUSION — Any Topic

FILL-IN-THE-BLANK

- " is not just an option but a necessity for a sustainable future."
- **Collective action vs collective suicide.** — Antonio Guterres
- **Saptrishi — Green Growth.**

Subtopic: Wetlands

Card 1: MANGROVES

INTRODUCTION

- **Definition:** Salt-tolerant vegetation that grows in **intertidal estuaries**.
- Key initiatives:
 - 1. **Mangroves for Future.**
 - 2. **Mangroves Atlas.**
- **Map:** Sunderbans, A&N, Cauvery, KG basin, Mahanadi basin, Ratnagiri, Goa, Kori Creek, Kutch.
- **SAMIE Initiative — Sustainable Aquaculture in Mangrove Ecosystem** — planting mangroves around shrimp ponds (no fish feed needed + active conservation).

DATA

- **ISFR 2019:** Mangrove cover in Indian subcontinent — **~5,000 sq km** (increased by **1.1%**).

Card 2: WETLANDS

INTRODUCTION

- **Definition (RAMSAR):** Areas of submerged lands — marshes, swamps, bogs, fens — usually up to **6m water depth**.
- **7 lakh wetlands** in India.
- **Deepor Beel** — **1/3rd encroached**.
- **Key Principle: Wise Use.**
- **Best Practice: Keshopur Maini Community Reserve.**
- **Way Forward:** Need for scientific criteria to identify wetlands + data bank + cooperative and decentralised efforts.

DATA

- In the last **100 years** — **65% of wetlands have disappeared**.
- **7 lakh** wetlands in India.

Section Group: Pollution & Degradation

Microtopic: Environmental pollution and degradation

Subtopic: Air Pollution and Related Issues

Card 1: Pollution Control & NCAP

QUOTE

"What we breathe and drink today decides who we become tomorrow."

INTRODUCTION

- Through the **National Clean Air Programme**, **Namami Gange**, and **Soil Health Cards**, India is tackling pollution in a mission mode—aligning with a “**right to life is a right to clean environment**” vision.

EXAMPLES

- **NCAP cities showed 25–30% PM2.5 reduction** in 5 years.
- + sewerage projects launched under **Namami Gange**.
- Soil health card scheme distributed **23 crore+ cards**.

CONCLUSION

- Pollution control needs **decentralised monitoring and behavioural change** for deep-rooted impact.

Card 2: AIR POLLUTION

INTRODUCTION

- India is the **5th most polluted country**.
- **WHO: 14 of 15** most polluted cities in the world are in India.
- **Steps taken:**
 - 1. NCAP.
 - 2. CAQM.
 - 3. EPCA.
 - 4. Smog towers.
 - 5. **Fly Ash Management Rules 2016** (India — only **63% fly ash utilisation**).
- **Best Practices:** London, Beijing models.
- **Transport sector** contribution:
 - 1. **23% of CO₂**.**
 - 2. ***3rd most CO₂-emitting sector.***
 - 3. ***90% emission from road transport.***
 - 4. **Vehicles expected to** double to 200 million by 2030.
 - 5. **Schemes** — FAME, BS-VI.**
- **Agriculture + livestock: 18% of GHG.**
- **Cities contribute 60% of GHGs and 70% of GDP.**

DATA

- **CSE:** Life expectancy in India decreased by **2.5 years** due to air pollution.
- As of 2020:
 - 1. **90% of TPPs** not compliant with sulphur standards.
 - 2. **>51%** not compliant with PM standards.

Subtopic: Mining and Erosion

Card 1: DESERTIFICATION

INTRODUCTION

- **Desertification** = deterioration of quality and productivity of land due to **climatic variation and human activity**.
- Key frameworks:
 - 1. **UNCCD**.
 - 2. **30×30 target**.
 - 3. **Bonn Challenge**.
- Govt schemes:
 - 1. **CAMPA**.
 - 2. **Green India Mission**.
 - 3. **Urban Forestry** — Pune Wajre Forest Model.**
 - 4. ****iForest Model**.
- International initiative: **Great Green Wall** — Sahel region.
- **Best Practice: Project BOLD / Bamboonomics** — Bamboo oasis on arid/semi-arid lands — tribal livelihood + erosion control.

DATA

- **30% of total land** threatened with degradation — **ISRO**.
- **MoEFCC: 2.5% of GDP** already lost due to desertification.
- Since 1850s — land under cultivation has become **3×**.

Card 2: LAND SUBSIDENCE

INTRODUCTION

- **Definition:** Shrinking of the ground due to underground material movement — gradual settling or sudden sinking.
- **Mishra Committee 1976** — key committee on land subsidence.
- **Best Practice: Vetiver system** for soil erosion control.

Subtopic: Solid wastes

Card 1: SOLID WASTE MANAGEMENT — General

INTRODUCTION

- **Waste generated:**
 - 1. **65 MT/year.****
 - 2. **Projected 165 MT by 2030.****
- Out of **63 MT**:
 - 1. **43 MT collected.****
 - 2. **Only 13 MT treated.****
- Principle: **Waste to Wealth.**
- **Best Practices (Waste Water):**
 - 1. Kolkata Sewage-Fed Aquaculture.
 - 2. Avadi Sewage Plant (TN).

DATA

- 1. **65 MT/yr** waste generated.**
- 2. **Projected 165 MT by 2030.****

Card 2: E-WASTE

INTRODUCTION

- **Definition:** All items of **electric and electronic equipment (EEE)** and their parts discarded after use.
- 1. India remains the **3rd largest e-waste generator (generating approx. 4.1 MT in 2023).**
- 2. Under **25% collected and recycled formally.**
- **Best Practices:**
 - 1. **LIAM (Apple recycling robot).****
 - 2. ***EU Right to Repair.***
 - 3. ***Bhopal eWaste Clinic.***
 - 4. ****EPR (Extended Producer Responsibility).**
- **Basel Convention:** Controls **transboundary movement** of hazardous waste including e-waste.

DATA

- India remains the **3rd largest e-waste generator (generating approx. 4.1 MT in 2023).**
- Under **25% collected and recycled formally.**

Card 3: PLASTIC POLLUTION

INTRODUCTION

- 1. India generates **9.3 MT/year** of plastic waste.
- 2. **43% is Single-Use Plastic (SUP).**
- Only **30% recycled**.
- China and Malaysia have **banned illegal plastic imports** from USA, Canada, UK.
- **Steps:**
 - 1. Plastic Waste Management Amendment Rules 2021.
 - 2. India Plastic Challenge Hackathon 2021.
 - 3. **Puneet Sagar Abhiyan.**
 - 4. **EPR on plastic packaging.**
- **Best Practices:**
 - 1. AP — Rice for Plastic.
 - 2. Ireland — **Plas Tax.**
 - 3. Precycling.
 - 4. South Korea reusables.
 - 5. **MP Bhartan Bhandar.**
 - 6. **Wayanad** — plastic in road construction.
 - 7. **Main Indoori Jholadhari.**

DATA

- 1. India — **9.3 MT/yr.**
- 2. **43% SUP.**
- **OECD Global Plastics Outlook** — plastic waste will **3× by 2060.**

Card 4: CIRCULAR ECONOMY

INTRODUCTION

- **Definition:** Economy where products are designed for **durability, reuse and recyclability.**
- **6R Framework: Reduce, Reuse, Recycle, Refurbish, Recover, Repair.**
- **Govt Steps:**
 - 1. Vehicle Scrappage Policy.
 - 2. Right to Repair Bill.
 - 3. National Productivity Council push.
 - 4. **India-Australia Circular Economy Hackathon (I-ACE).**
 - 5. Digital India (e-waste recycling component).
- **International:**
 - 1. China — Circular Economy Promotion Law.
 - 2. Germany & Japan — binding principles.
 - 3. **SDG 12.**

CONCLUSION

- Shift from **"Take, Make, Dispose"** mindset to **6R** approach.

Card 5: BIO-MEDICAL WASTE

INTRODUCTION

- **Definition:**
 - 1. Human and animal anatomical waste.
 - 2. Treatment apparatus like needles, syringes etc.
- **Colour-coded categorisation:** Blue, Red, Yellow, White.
- Governed by **BMW Rules 2016**.
- Only **15% of BMW is hazardous** (in general).

DATA

- **Drastic increase during COVID: 20 T/day → 200 T/day — CPCB.**

Subtopic: Water Pollution and Related Issues

Card 1: COASTAL ECOSYSTEMS & EROSION

INTRODUCTION

- Indian coastline:
 - 1. **7,500 km + 1,200+ islands.**
 - 2. **1/5th of population + 3 metro cities.**
- **Coastal Erosion** = wearing away of land by high winds, drainage, wave action, tidal currents.
- **Kerala Coastal Zone Management Project** — integrated coastal zone management for erosion, sustainable development, conservation.
- Technology: **Geo-textiles** for stabilisation.
- **Climate Action Plan:** Detailed framework for measuring, planning and reducing **GHG emissions** + preventive measures.
- **BMC** drafting **Mumbai Climate Action Plan (MCAP)**.
- Govt steps:
 - 1. **NAPCC.**
 - 2. **CRZ.**
 - 3. **National Coastal Mission.**

CONCLUSION

- Impact of **climate change on coastal areas** has strong **strategic, economic and political repercussions.**

DATA

- **MoEFCC: 40%** of Indian coastline subjected to **high, medium or low coastal erosion.**

Card 2: WATER — General Overview

QUOTE

"We never know the worth of water until the well is dry." — Thomas Fuller

DATA

- **NITI Aayog:** Demand will outstrip supply by 2030.

INTRODUCTION

- India has only **4% of world's freshwater** but **17% of population**.
- **Per capita availability:**
 - 1. 1951 — **5177 m³**.
 - 2. 2011 — **1545 m³**.
 - 3. 2030 projected — **1300 m³**.
- **70% of water contaminated** with heavy metals.
- Pakistan — increased **domestic violence** due to water insecurity & heat-led aggression.
- Sahel — conflicts due to reducing fertility from **water stress**.
- Social issues: **Water wives** phenomenon.
- **Govt steps:**
 - 1. Jal Shakti Mission.
 - 2. MGNREGA (recharging aquifers).
 - 3. **Atal Bhujal**.
 - 4. River Interlinking.
 - 5. Water Act.

Card 3: GROUNDWATER

INTRODUCTION

- **WB:** India's groundwater depleting at **0.4 m/year**.
- Caters to **80% of rural domestic water needs** and **50% of urban water needs**.
- **90% used for agriculture**.
- **Govt efforts:** **CGWB / CGWA**.

DATA

- India's groundwater depleting at **0.4 m/year** — World Bank.

Card 4: JAL JEEVAN MISSION

INTRODUCTION

- Budget allocation increased by **23%**.
- **WHO:** **4 lakh diarrhoea deaths** can be averted if JJM meets its target.

DATA

- Budget allocation up by **23%**.
- **4 lakh diarrhoea deaths** preventable — WHO.

Card 5: EUTROPHICATION

INTRODUCTION

- **Definition:** Excessive plant and algal growth due to increased availability of limiting growth factors — **sunlight, CO₂ and nutrients** needed for photosynthesis.

Card 6: MARINE LITTER

INTRODUCTION

- **8 million tonnes** of plastic enter oceans every year.
- Plastic constitutes **80% of all marine debris**.
- **Bioaccumulation & biomagnification** — microbeads found in blood of newborns.
- **Indian Efforts:**
 - 1. Plastic Waste Management Rules 2018.
 - 2. **KL's Suchitwa Mission** (fishermen collecting plastic).
 - 3. **Puneet Sagar Abhiyan**.
- **International Efforts:**
 - 1. **GloLitter Partnership Project**.**
 - 2. **Ocean Clean-Up**.
 - 3. ***MARPOL by IMO.***
 - 4. ****SDG 14**.
- **Cocos Keeling Island** — example of marine litter impact.

DATA

- **8 million tonnes** of plastic end up in ocean every year.
- Plastic = **80%** of all marine debris.

| DISASTER MANAGEMENT

Section Group: Frameworks & Preparedness

Microtopic: Disaster Management: Preparedness, resilience, and vulnerability frameworks

Subtopic: Disaster Preparedness and Resilience

Card 1: DISASTER MANAGEMENT — TEMPLATES (General)

INTRODUCTION

- **Impact dimensions:**
 - 1. Socio-economic.
 - 2. Infrastructure.
 - 3. Environment.
 - 4. Epidemics & pandemics.
 - 5. Psychological.
 - 6. Political/Governance.
- **Indian Framework:**
- **DM Act** — 3-tier structure:
 - 1. **NDMA.**
 - 2. **SDMA.**
 - 3. **DDMA.**
- **NDMP 2016** — aligns India with **Sendai Framework**.
- 1. **NDRF.**
- 2. **National Disaster Response Fund.**
- 3. **National Calamity Contingent Duty.**
- **SENDAI Framework** — Reduce 4, Increase 3:
- Reduce:
 - 1. Mortality.
 - 2. Affected people.
 - 3. Economic loss.
 - 4. Damage to critical infra.
- Increase:
 - 1. Countries with DRR strategies.
 - 2. International cooperation.
 - 3. Access to multi-hazard EWS.
- **Keywords:**
 - 1. **DRR, Vulnerability mapping, Prevention & Mitigation, Emergency response, L&D, Local genius, First responders, 3R's (Relief.**
 - 2. **Rehabilitation.**
 - 3. **Reconstruction), BBB (Build Back Better).**

CONCLUSION

- **"Disasters are inevitable, but their impact is not."** — through robust DM, we can mitigate risks, save lives and build resilient communities.
- **Tehri Dam case** — SC: DM is part of the **right to sustainable development**.
- **"Prepare and prevent, not repair and repent"** — must become universally engrained.

Card 2: EXTREME WEATHER EVENTS (EWEs)

INTRODUCTION

- **IPCC Definition:** Events that are **rare at a particular place and time**.

DATA

- **CSE Report 2022:** EWEs recorded on **241/273 days** of the year.
- India recorded warmest, driest and wettest days in **2022**.
- **75% of districts** affected with EWEs.

Card 3: DM INNOVATIVE SUGGESTIONS

INTRODUCTION

- 1. **Safety culture.****
- 2. **Decentralised, pro-active, preventive, community resilience, self-sufficiency.**
- 3. **Parallel rather than sequential approach.**
- 4. ****equitable.**
- **Funding:** Catastrophe bonds.
- **Resilient Infra:**
 - 1. SDG 9.
 - 2. **CDRI.**
- **EWS:** OD SATARK Mobile App.
- **WF mnemonic — RW wants LES VAT LSD:**
 - 1. **R** — Risk coverage.
 - 2. **W** — Women role.
 - 3. **L** — Leadership.
 - 4. **E** — EWS.
 - 5. **S** — Sendai.
 - 6. **V** — Vulnerability mapping.
 - 7. **A** — Academia.
 - 8. **T** — Tech.
 - 9. **L** — Local capacity.
 - 10. **S** — Social media.
 - 11. **D** — DRR.

CONCLUSION

- **"Every \$1 spent on mitigation saves \$7 on response and recovery"** — Sendai.
- **Zero Casualty Approach.**
- **Tehri Dam Case** — SC: DM is part of right to **sustainable development.**

DATA

- Mitigation ROI: **\$1 → \$7 saved.**

Card 4: Template 9: DISASTER MANAGEMENT — General Template

FILL-IN-THE-BLANK INTRO (IMPACT SECTIONS)

- Socio-economic / Infrastructure / Environment / Epidemics & pandemics / Psychological / Political & governance.

FILL-IN-THE-BLANK CONCLUSION

- **"Disasters are inevitable, but their impact is not.** Through robust disaster management strategies, we can mitigate risks, save lives and build more resilient communities."

WF MNEMONIC — RW WANTS LES VAT LSD

- 1. **R** — Risk coverage.
- 2. **W** — Women role.
- 3. **L** — Leadership.
- 4. **E** — EWS.
- 5. **S** — Sendai.
- 6. **V** — Vulnerability mapping.
- 7. **A** — Academia.
- 8. **T** — Tech.
- 9. **L** — Local capacity.
- 10. **S** — Social media.
- 11. **D** — DRR.

Subtopic: Vulnerability and Risk Assessment

Card 1: DISASTER — Definition & Vulnerability

INTRODUCTION

- **Definition:** Disruption on a massive scale (natural or man-made), occurring in short or long period — **affects the capacity of society to cope with its own resources.**

DATA

- India vulnerable to **30 different types of disasters.**
- 1. ****EQ** — 60%.
- 2. Cyclones — 12%.
- 3. Floods — 12%.
- 4. Drought — 68%** of India vulnerable.
- **5700/7500 km** coastline prone to cyclone & tsunami.

Section Group: Specific Disasters & Hazards

Microtopic: Disasters: Specific natural and man-made hazards

Subtopic: Cloudburst

Card 1: CLOUDBURSTS

INTRODUCTION

- **Definition:** Short-duration intense rainfall — **100 mm/hour** over a small area.
- **Best Practice:**
 - 1. Copenhagen Cloudburst Formula — blue-green infrastructure.
 - 2. City planning.
- **Hotspot: CNP (Cherrapunji/Northeast)** — world's most vulnerable spot for severe thunderstorm formation.

DATA

- Intensity: **100 mm/hr** over small area.
- Examples:
 - 1. UK 2021.
 - 2. JK 2010.
- Vulnerable terrains: mountains, deserts, interiors of landmass.

Subtopic: Cyclone

Card 1: BOMB CYCLONE / BLIZZARD / SNOW STORM

INTRODUCTION

- **Bomb Cyclone (Bombogenesis):** Rapidly strengthening storm where **pressure drops 24 mb in 24 hours**.

Card 2: CYCLONES

INTRODUCTION

- **Definition:**
 - 1. Rapid inward air circulation around a **low-pressure area** — perennial phenomenon in India.
 - 2. **"silent killer"**.
- **IMD** — colour-coded warnings.
- **OD (Odisha) Model** — best practice in cyclone preparedness.
- **RIMES** — Regional Integrated Multi-Hazard Early Warning System.
- **Conditions:**
 - 1. Temp >27°C.
 - 2. Coriolis force.
 - 3. Small variation in vertical wind speed.
 - 4. Low pressure.
 - 5. Upper air divergence.
- **Why increasing in Arabian Sea:**
 - 1. IOD heating.
 - 2. GHG.
 - 3. El Niño.
 - 4. Weakening monsoon.
 - 5. Low wind shear due to CO₂.
- **Multiple engagement areas:**
 - 1. Natural storm breaks (mangroves).
 - 2. Tech (UAV, satellite).
 - 3. Coordination.

DATA

- India accounts for **10% of world's tropical cyclones**.
- **13 states and UTs** affected.
- Cost: **2% of GDP**.
- **Cyclone Nivar**: damaged **23,000 acres** of crop.

Card 3: STORM SURGE

INTRODUCTION

- **Definition:** **Abnormal rise of sea level** as cyclone crosses the coast, leading to inundation of coastal strip.

Card 4: TEMPERATE CYCLONE

INTRODUCTION

- **Definition:** Cyclones developing in **middle and high latitudes (35°–65° N&S)** beyond the tropics.
- Explained by **Polar Front Theory**.

Subtopic: Drought

Card 1: DROUGHT

INTRODUCTION

- **Definition:** Period of **abnormally dry weather** causing serious **hydrological imbalances** in the area.
- **Note:** Not defined as disaster under the **DM Act**.
- **Drought Crisis Management Plan 2015.**
- **Govt steps:**
 - 1. MGNREGA (employment).
 - 2. Recharge groundwater.
 - 3. Crop loss compensation.
- **Schemes:**
 - 1. PMGSY.
 - 2. Soil Health Card.
 - 3. **PM Fasal Bima Yojana.**
- **4 Types:**
 - 1. Meteorological.
 - 2. Hydrological.
 - 3. Agricultural.
 - 4. Ecological.

DATA

- **70%** of India drought-prone.
- **20% of India's land** facing drought-like conditions.

Card 2: FLASH DROUGHT

INTRODUCTION

- **Definition:** **Rapid and intense onset** of drought over a **few weeks or months** — unlike traditional drought. Caused by: high temperatures, reduced precipitation, low soil moisture, increased evaporation.
- **Best Practices:**
 - 1. **Seechwal Model.****
 - 2. ****Israel case study.****
 - 3. ****Agro-climatic agriculture.**

DATA

- Study: India could experience **more flash droughts by end of century.**

Subtopic: Earthquake

Card 1: EARTHQUAKES

INTRODUCTION

- **Definition:** Shaking of Earth's surface caused by **seismic waves** from sudden release of energy in the **Earth's crust or upper mantle**.
- **Way Forward:** EIA before mining and dam construction.
- **Best Practices:**
 - 1. **Bhungas (Kutch).****
 - 2. ****Ikra Bamboo Houses (Sikkim).**
- **Steps:**
 - 1. National Retrofit Program 2014.
 - 2. National EQ Risk Mitigation Project.
 - 3. **EQ Disaster Risk Indexing.**
- **Best Practice (International):** Japan — **80% compliance** with quake-resistant buildings.

DATA

- Major EQs:
 - 1. GJ 2001.
 - 2. Nepal 2015.
 - 3. SK 2011.
- **60% of India's land** vulnerable (**80% of population**).

Card 2: EARTHQUAKE SWARMS

INTRODUCTION

- **Definition:**
 - 1. Series of low-magnitude earthquakes in a localised region over **days to months** — **no clear foreshock/main shock/aftershock sequence**.
 - 2. **Difficult to predict.**
- **Causes:**
 - 1. Hydro-seismicity.
 - 2. Magnetic activity.
 - 3. Weak crust.
- **Example:** Palghar, MH — **30 low-intensity EQs in 2 months.**

Subtopic: Landslide

Card 1: LANDSLIDES

INTRODUCTION

- **Definition:** Downward and outward movement of slope material (rock, debris, earth) under **gravity**.
- **Way Forward:**
 - 1. NDMA — **NLRMS** (National Landslide Risk Mitigation Strategy).
 - 2. National Landslide Risk Mitigation Project.
- **Steps:**
 - 1. GSI hazard zonation maps using UAV.
 - 2. **Microzonation**.
 - 3. Capacity building.
- **Best Practices:**
 - 1. MH — **geo-textile**.
 - 2. Uttarkashi — **EWS + stabilisation**.
- **70 HEPs in Himalayan states** — EQ-triggered landslide risk.

DATA

- **GSI: 12% of land area** prone to landslides.
- Vulnerable areas: **Himalayas + NE + Western Ghats**.
- **182 landslides** across 10 states in 2022.

Subtopic: Miscellaneous (Disasters)

Card 1: HEATWAVES

INTRODUCTION

- **Definition:** Period of **abnormally high temperature** — fatal to humans and animals.
- **Way Forward:**
 - 1. NDMA guidelines.
 - 2. **Sensitive approach** — identifying most vulnerable.
- **Best Practices:**
 - 1. **Nagpur** — Heat Action Plan + Heat Wave Index.
 - 2. **Ahmedabad** — Cool Roofs Program for slums + colour-coded heat alerts.

DATA

- **IMD:**
 - 1. >40°C plains.
 - 2. >27°C coastal.
 - 3. >30°C hills + 5–6° above normal.
- **2015 New Delhi heat wave:** 1,800 died.
- **2022 — Europe, N. America, China:** >20,000 died.
- **WB:**
 - 1. **75% of Indian labour force** is heat-exposed.**
 - 2. **Situation could break** human survivability limit.**

Card 2: COLD WAVE

INTRODUCTION

- **Definition:** Rapid fall in temperature within **24 hours** requiring substantially higher protection for agriculture, industries and social activities.
- **Best Practice: Rein Basher** (night shelters) in Delhi.
- Prepare an **action plan** for cold wave management.

DATA

- Conditions: Plains < **10°C + 4.5° below normal** for 2 consecutive days.
- **IMD: 3× increase** in number of cold wave days (2017–20).

Card 3: VOLCANOES

INTRODUCTION

- **Definition:** Opening in the Earth's crust from which **molten rock (magma), gases and debris** erupt.
- **Positive effects:**
 - 1. Soil fertility.
 - 2. Geothermal energy & hot springs.
 - 3. Minerals.
 - 4. **Cooling effect.**

Card 4: FOREST FIRES

INTRODUCTION

- **Definition:** Major fires occurring **unpredictably** in combustible environments (dry forests), burning for **long term over large area**.
- **Steps:**
 - 1. **National Action Plan on Forest Fires 2018.****
 - 2. ***FF Alert System.***
 - 3. ****i-Forest initiative.**

DATA

- **FSI:** **36%** of forests vulnerable.
- Example: Kullu Valley forest fire 2021.
- **FSI:** Forest fire cost — **\$1.5 Bn.**

Card 5: FLOODS

INTRODUCTION

- **Definition:** Situation of high water level along a river or coastal areas leading to **inundation of land**.
- **NDMA guidelines:** Structural (reservoirs, desilting, afforestation) + Non-structural (**flood plain zoning, EWS**).
- **Best Practices:**
 - 1. Buoyant buildings (Netherlands).
 - 2. Bamboo houses (NE India).
 - 3. ****Rain gardens.**
 - 4. Sponge Cities.**
 - 5. Jabalpur — **rebate on property tax for RWH.**

DATA

- **National Flood Commission:**
 - 1. **40 mn ha** prone to floods.**
 - 2. **Cost \$7 Bn.****
- **80% rain in 3 months** + cloudbursts.
- **12% of India** flood-vulnerable.
- India: **1/5th of global flood deaths.**

Card 6: LIGHTNING STRIKES

INTRODUCTION

- **LRIC** (Lightning Resilient India Campaign) — IMD, NDMA and others — aim: reduce deaths to **<1,200/annum**.
- **"Damini" app** — IITM Pune — **3-hour advanced warning**.

DATA

- **1.85 crore lightning strikes** recorded between 2020–21.

Card 7: GLOF (Glacial Lake Outburst Flood)

INTRODUCTION

- **Definition:** Breach of **glacial moraine wall** causing flooding due to increased water content exceeding capacity.
- **Vulnerability:**
 - 1. India — "3rd pole".
 - 2. **8,800 glacial lakes**.
 - 3. CC.
 - 4. Tectonically active.

DATA

- **2013** — Kedarnath tragedy.
- **2021** — UK Chamoli disaster.
- **8,800 glacial lakes** in India.

Card 8: NUCLEAR DISASTERS

INTRODUCTION

- Governed by **Public Liability Insurance Act**.
- **Vulnerability:**
 - 1. 22 functioning nuclear reactors.
 - 2. Shares boundary with **2 nuclear states**.
 - 3. Reactors on coastline (**Kalpakkam NPP flooded by 2004 tsunami**).

DATA

- **22 functioning nuclear reactors** in India.

Subtopic: Tsunami

Card 1: TSUNAMI

INTRODUCTION

- **Definition:** Series of **extremely long waves** caused by large and sudden displacement of ocean water due to EQ/volcanoes.
- **Way Forward: DOARS** (Deep Ocean Assessment and Reporting System).
- **INCOIS** (by MoES) — set up **ITWEC** (Indian Tsunami Early Warning Centre).
- **Sagar Vani app** — community alert.

DATA

- **2004 Tsunami: 2.3 lakh people died.**

Subtopic: Urban floods

Card 1: URBAN FLOODS

INTRODUCTION

- **Definition:** Inundation of **urban infrastructure** due to intense rainfall overwhelming **drainage capacity**.
- **iFlows Mumbai** — integrated flood management system.
- **RWH (Rainwater Harvesting)** as key solution.
- **Germany:** Sponge Cities — **Hamburg model**.

INTERNAL SECURITY

Section Group: Border Management & Organized Crime

Microtopic: Linkages of organized crime with terrorism.

Subtopic: Organized Crime and Terrorism

Card 1: RADICALISATION

INTRODUCTION

- **Definition:** Process by which an individual comes to adopt **extreme political or religious ideas & aspirations**.
- **Best Practices:**
 - 1. **Aqaba Process (Jordan).**
 - 2. **Maharashtra Model** — 3-pronged counselling (family + clergy + professionals) + reintegration.

Card 2: NEW FORMS OF MILITANCY

INTRODUCTION

- **Over-Ground Workers (OGW)** — help militants in movement, ammunition supply & accommodation.
- **Hybrid militants** — not listed as ultras but radicalised enough to carry out terror strikes, then slip back to routine life (sleeper cells).
- **Keywords:**
 - 1. **Faceless militancy.**
 - 2. **Mission Pehal.**
 - 3. **Lone wolf.**
- **Best Practice: SL Restorative Justice Model** — vocational, spiritual, social & psychological rehabilitation of LTTE militants into mainstream.

Card 3: ORGANISED CRIME

INTRODUCTION

- **Definition:** **Coordinated criminal activities** by structured groups for financial gain.
- New Bills replacing IPC, CrPC have separate provisions for organised crime — step in the right direction.

Card 4: TERRORISM

INTRODUCTION

- **Definition:** Illegal use of **violence or threats** to spread fear among civilians or governments for **political or ideological goals**.
- India has been a **victim for decades**, esp. **state-sponsored terrorism**.
- **UK's CONTEST Strategy — 4Ps:**
 1. **Prevent** — counter radicalisation.
 2. **Pursue** — detect, disrupt & prosecute (e.g., NATGRID, NIA).
 3. **Protect** — access restriction, border security, cyber (e.g., CIBMS).
 4. **Prepare** — resilience (e.g., NSG).
- Others:
 1. **UN CCIT.**
 2. **FATF.**

CONCLUSION

- In light of increasing tech sophistication, **"War on Terror" demands beyond-border cooperation, suffocating terror finance, building resilient societies & robust institutions.**

Card 5: LINKAGES BETWEEN ORGANISED CRIME & TERRORISM

INTRODUCTION

- **Convergence** of OC & terrorism through **resource sharing and mutual facilitation** poses significant threats to **internal security and global stability**.

CONCLUSION

- Template — the **war is not that of any one agency or one govt alone.**

Card 6: TERROR FINANCE

INTRODUCTION

- **Definition:** Provision of funds/financial support to individual terrorists or groups.
- Recently **"No Money for Terror"** conference held in **New Delhi**.
- **Steps:**
 1. UAPA Amendments.
 2. Crypto KYC.
 3. Dedicated cell in NIA.
 4. **Soft Demonetisation.**
 5. FCRA amended (NGOs).
 6. **FATF + PMLA.**

CONCLUSION

- **Suffocating finance** is necessary but not sufficient to tackle the **hydra-headed menace of terrorism.**

Microtopic: Security challenges and their management in border areas

Subtopic: Border Area Management

Card 1: BORDER SECURITY — General

QUOTE

"Good fences make good neighbours." — Robert Frost

CONCLUSION

- Need to view border security through a **holistic prism** of vibrant **P2P exchanges**, upgraded surveillance, and **robust bilateral cooperation**.

DATA

- Land border: **>15,000 km with 7 countries**.

INTRODUCTION

- India has a **land border of >15,000 km with 7 countries**.
- Each border poses **unique challenges** for internal security.

Card 2: CHINA BORDER

INTRODUCTION

- India shares **3,488 km** of border with China — remains **disputed at multiple locations**.
- Example: Recent **Ladakh standoff**.
- **Vibrant Village Program** — initiative to develop border villages.
- Issues: **Cartographic & Nomenclatural aggression**.

DATA

- **3,488 km** India-China border.

Card 3: NEPAL & BHUTAN BORDER

INTRODUCTION

- **Open border** with Nepal — no travel restrictions for citizens on either side.
- Bhutan border — similar open-border arrangement.

Card 4: MYANMAR BORDER

INTRODUCTION

- India-Myanmar border runs along **4 states in Northeast India**.
- Marked by **Free Movement Regime (FMR)** — up to **16 km on either side** due to deep ethnic and cultural ties.

DATA

- FMR extends **16 km** on either side.

Card 5: PAKISTAN BORDER

INTRODUCTION

- Pakistan — India's **northwestern neighbor** sharing maritime & land boundary demarcated by the **Radcliffe Award of 1947**.
- **Madhukar Gupta Committee** — reviewed border management.
- Measures:
 - 1. **Village Defence Guards**.
 - 2. **CIBMS** (Comprehensive Integrated Border Management System).

Card 6: BANGLADESH BORDER

INTRODUCTION

- India shares a **long porous land, riverine & maritime border** with Bangladesh along India's **east and northeast region**.

Card 7: Template 3: BORDER SECURITY — ___ Poses Unique Challenges

QUOTE

"Good fences make good neighbours." — Robert Frost

Subtopic: Maritime Security challenges

Card 1: MARITIME SECURITY

INTRODUCTION

- India has a coastline of **7,517 km** and **>1,200 islands** — highly vulnerable to maritime threats.
- India's vision: **"Neel Kranti se Arth Kranti"**.
- Rising threats from **China in IOR**.

CONCLUSION

- Need for **building domestic capacity** along with **International push** — Russia's proposal for a dedicated maritime security body in **UN Security Council**.

DATA

- Coastline: **7,517 km + >1,200 islands**.

Card 2: PIRACY

INTRODUCTION

- **Definition:** Criminal acts like **robbery, hijacking, or hostage-taking** targeting ships in international waters.

Section Group: Cyber, Financial & Media Security

Microtopic: Challenges to internal security through communication networks, role of media and social networking sites, and basics of cyber security

Subtopic: Cyber security

Card 1: Cybersecurity & DPDP Act

QUOTE

"A secure cyberspace is non-negotiable in a digital republic."

INTRODUCTION

- The **CERT-In directives** and **Digital Personal Data Protection Act (DPDPA)** signify India's resolve to safeguard digital sovereignty, anchoring the vision of a **"cyber-resilient Bharat"** amid rising global threats.

EXAMPLES

- Mandatory breach reporting within **6 hours** under CERT-In norms.
- DPDPA established **Data Fiduciary & Principal** framework.
- **Cyber Surakshit Bharat** trained 300+ officers in data governance.

CONCLUSION

- India must ensure **capacity-building and compliance** to make data protection both enforceable and equitable.

Card 2: CYBER SECURITY

QUOTE

"I dream of a digital India where cyber security becomes an integral part of national security."

INTRODUCTION

- **Definition:** Securing **digital systems, networks and data** from cyber threats to ensure **confidentiality & integrity**.

DATA

- Internet Security Threat Report: India **3rd most vulnerable** to cyber threats.
- **NCRB 2024 records a 17% surge in cybercrimes**, crossing 1 lakh reported cases (over 72% being financial fraud).

Subtopic: Digital Media and Security Threats

Card 1: FAKE NEWS

INTRODUCTION

- **Definition:** Information deliberately **fabricated and aimed at deceiving** others into believing falsehood.
- **Best Practice:** **KL Model** of school education for media literacy.

Card 2: ROLE OF MEDIA & SOCIAL NETWORKING SITES

QUOTE

"The new frontiers of war is the civil society which can be manipulated to hurt a nation's interests." — Ajit Doval

Subtopic: Mob Violence

Card 1: MOB LYNCHING

INTRODUCTION

- **Definition:** Extra-judicial killing of a person by **mob action** (e.g., Udaipur, Una flogging).
- SC recently sought updates on action taken by govt.
- **Steps:**
 - 1. IT Rules 2021, 2023.
 - 2. Amendment to IPC, CrPC.
 - 3. **RJ Protection from Lynching Bill 2019.**
 - 4. **UAPA amendments.**
- **Tehseen Poonawala case** — duty of state to protect citizens & "**pluralistic social fabric**" of the country.

Microtopic: Money-laundering and its prevention, including illicit trafficking

Subtopic: Drug trafficking

Card 1: DRUG TRAFFICKING & ABUSE

INTRODUCTION

- India's location between the **Golden Triangle & Golden Crescent** + porous borders = highly vulnerable to drug trafficking.
- **Steps:**
 - 1. **Nasha Mukht Bharat.**
 - 2. **NIDAAN portal.**
 - 3. **NAPDDR** (National Action Plan for Drug Demand Reduction).
 - 4. **NDPS Act & NCB.**
- **Operation Samudragupta** — NCB & Navy seizure of **₹15,000 crore** in Arabian Sea.
- **Best Practice: Mansa Village (PB)** — diverting youth energy into community work — "Mera Pind Mera Maan".
- **3-pronged reduction:** Demand + Supply + Harm.

CONCLUSION

- State is duty bound under **Article 47** to prohibit consumption of drugs to safeguard **social fabric, public health & internal security.**

DATA

- Operation Samudragupta: **₹15,000 crore** seized.

Subtopic: Money-laundering and Prevention

Card 1: MONEY LAUNDERING

QUOTE

"Money laundering is a heinous crime which not only affects the socio-economic fabric of a nation but promotes other heinous crimes." — Vijay Mohanlal Choudhary vs UOI

DATA

- **IMF: 2–5% of world GDP** is laundered annually.

INTRODUCTION

- **Definition:** Process by which illegally obtained money is made to appear as if it originated from a **legitimate source**.
- **Stages:** Placement → Layering → Integration.
- **Steps:**
 - 1. **PMLA Rules 2023** (stringent KYC).
 - 2. **Benami Transactions Prohibition Act**.
 - 3. **PMLA Act 2003** (cognisable & non-bailable + confiscation).
 - 4. **Black Money Act**.
 - 5. Deregistered shell companies.

Card 2: FATF

INTRODUCTION

- **FATF** — Global watchdog that ensures **international cooperation** and sets **international standards** to tackle **ML & TF** (Money Laundering & Terrorist Financing).

Section Group: Extremism & External Threats

Microtopic: Linkages between Development & spread of Extremism

Subtopic: Left-Wing Extremism

Card 1: LEFT WING EXTREMISM (LWE / NAXALISM)

QUOTE

"Underdevelopment and inequality — hence need to imbibe Tribal Panchsheel principles in bringing peace & participative governance." — Antonio Guterres

CONCLUSION

- **53–55% reduction** in casualties.

DATA

- **53–55% reduction** in LWE casualties.

INTRODUCTION

- **Definition:** Radical ideologies advocating **socio-political change through armed struggle** against established authorities.
- **Steps:**
 - 1. Aspirational Districts.
 - 2. Road & Mobile tower connectivity.
 - 3. **PESA, FRA, Van Dhan.**
 - 4. Jal Jangal Jameen.
 - 5. Surrender & rehab project.
 - 6. Joint Task Force for interstate coordination.
- **SAMADHAN Doctrine:**
 - **S** — Smart leadership.
 - **A** — Aggressive strategy (fortified police stations).
 - **M** — Motivation & training (Greyhounds).
 - **A** — Actionable intelligence.
 - **D** — Dashboard-based KPIs.
 - **H** — Harnessing Tech (UAVs, biometric smart guns).
 - **A** — Action plan for each theatre.
 - **N** — No access to finance (PMLA).

Card 2: EXTREMISM

INTRODUCTION

- **Definition:** Advocacy of extreme ideologies or actions that challenge **established norms and institutions** through **unconstitutional & violent methods**.
- Antonio Guterres quote on underdevelopment.

Card 3: LINKAGES BETWEEN DEVELOPMENT & EXTREMISM

QUOTE

"Poverty is the mother of crime." — Marcus Aurelius

"The battle against extremism begins with addressing the roots of underdevelopment and inequality." — Antonio Guterres

Subtopic: N-E Insurgency

Card 1: NE INSURGENCY

INTRODUCTION

- **Definition:** Ongoing armed conflicts and movements seeking **political and social change** in India's northeastern states.

CONCLUSION

- Peace in NE is key to safeguarding **SUSI** of India as well as achieving PM's vision of **"Act East, Act first through North East"**.
- Antonio Guterres quote on underdevelopment and inequality.

Microtopic: Role of external state and non-state actors in creating challenges to internal security.

Subtopic: Threat from External state and Non-state actors

Card 1: HYBRID WARFARE

INTRODUCTION

- **Definition:** Use of **unconventional methods** as part of multi-domain operations **without engaging in open hostilities**.
- Examples:
 1. Chinese apps harvesting sensitive info.
 2. Lone wolf attacks.
 3. Cyber warfare.

Card 2: ROLE OF EXTERNAL STATE & NON-STATE ACTORS

INTRODUCTION

- **State actors** = foreign govts indulging in terrorism against another state (e.g., **ISI, Syrian war**).
- **Non-state actors** = extended arms of the state providing **plausible deniability** (e.g., **JeM, LeT** w.r.t. ISI).
- Through conventional & unconventional warfare, ___ has been posing **multifaceted security challenges** for India.

Card 3: Template 4: EXTERNAL STATE & NON-STATE ACTORS — ___ Posing Challenges

FILL-IN-THE-BLANK

- "Through conventional & unconventional warfare, ___ has been posing **multifaceted security challenges** for India."

Section Group: Security Forces & Mandates

Microtopic: Various Security forces and agencies and their mandate.

Subtopic: Act based

Card 1: IPC / CrPC / IEA CHANGES (3 New Bills)

INTRODUCTION

- **MoHA** introduced 3 new bills:
- IPC → **Bhartiya Nyay Sanhita Bill**.
- CrPC → **Bhartiya Nagrik Suraksha Sanhita Bill**.
- IEA → **Bhartiya Sakshya Bill**.
- New provisions — **Retributive, victim-centric, tech-enabled, simplified**.
- **IPC new sections:**
 - 1. S.110 (org crime).
 - 2. S.111 (terrorism).
 - 3. Cyber crimes.
 - 4. **Mob lynching — 7 years**.
 - 5. Police accountability.
- **CrPC changes:**
 - 1. Zero FIR.
 - 2. Mandatory videography.
 - 3. Timelines — chargesheet (90 days) & judgement (30 days).
 - 4. Compulsory forensics.
 - 5. Community service.
- **IEA changes:** Admissibility of **electronic/digital records**.
- **Issues:**
 - 1. Hate speech removed.
 - 2. Detention time increased.
 - 3. Low bail focus.
 - 4. Vague terms.

CONCLUSION

- **MoHA:** "The spirit of the 3 new laws is not to punish but to give justice to every citizen & prevent further crimes."

Subtopic: Mandates of Security forces

Card 1: GENERAL INTERNAL SECURITY CHALLENGES

INTRODUCTION

- India is **highly vulnerable** to security threats due to its location at the **epicenter of terrorism & instability**, as a **drug transit nation** with long vulnerable coastline and **porous and contested borders**.
- Solutions mnemonic — **SECURITY**:
- S — Security forces capacity building, fencing.
- E — Econ/Finance.
- C — Cooperation (bilateral/intl).
- U — Upgraded surveillance.
- R — Regional development / Good governance.
- I — Intelligence (locals), Infra, PPP.
- T — Trade/tech.
- Y — Youth / WHAM.

CONCLUSION

- Chanakya** — **Need 3 S** for holistic national power: **Shastra** (military prowess), **Shaastra** (academia/civil-military fusion), **Sanskriti** (ideals & principles).
- Multi-domain operations / Whole of Govt & society approach.**
- The fight against ___ requires **all nations & stakeholders** to come on the same platform.

Card 2: OVERALL DEFENCE

QUOTE

"Need National Security Strategy, National Defense Strategy, Higher Defense Organization in place." — General Manoj Narawane

KEYWORDS

- 1. Critical deficiencies.
- 2. Force multiplier.
- 3. Combat edge.
- Defence exports: \$1.5 billion** vs global market of **\$500 billion**.

DATA

- Defence exports: **\$1.5 Bn** vs global **\$500 Bn** market.

INTRODUCTION

- India's defence budget is the **5th largest in the world**.

Card 3: MILITARY REFORMS

INTRODUCTION

- Recent spate of reforms to improve **synergy, operational readiness & combat edge** of Indian Armed Forces.

CONCLUSION

- Move towards a **comprehensive National Security Strategy** encompassing **Chanakya's 3 S** — Shastra, Shaastra, Sanskriti. **Implementation will be key.**

Card 4: NIA (National Investigation Agency)

QUOTE

"The implementation is more important than their legislation."

INTRODUCTION

- NIA** is the **Central Counter-Terrorism Law Enforcement Agency** of India.
- Strengthen NIA w.r.t. **4 F** for protecting **SUSI**.

Card 5: CDS (Chief of Defence Staff)

INTRODUCTION

- CDS** is the **highest-ranking military officer** in India — responsible for coordinating and streamlining **overall military strategy and operations**.
- >70 countries** have a CDS equivalent.
- Roles:**
 - 1. Principal military advisor.
 - 2. Chairman COSC.
 - 3. Head of Dept of Military Affairs.
 - 4. Member of **DAC & Defence Planning Committee**.
 - 5. Advisor to **NCA (nuclear)**.
- Based on **Kargil Review Committee + Shekatkar Committee** recommendations.

CONCLUSION

- The **21st-century security challenges of a two-front war** call for a proactive CDS role in bringing **synergy, jointness & integration** + a cohesive **National Security Strategy**.

DATA

- >70 countries** have a CDS equivalent.

Card 6: INTEGRATED THEATRE COMMAND

INTRODUCTION

- **Definition:** Synergising capabilities & potential of **3 services** for **greater integration & resource optimisation**.
- Recommended by **Kargil Review Committee**.

CONCLUSION

- Will be effective only if supplemented with **military modernisation** and a **comprehensive National Security Strategy**.

Card 7: AGNIPATH SCHEME

INTRODUCTION

- Govt unveiled **Agnipath Scheme** to recruit youth into armed forces for a **short duration of 4 years**.

DATA

- Duration: **4 years**.

Card 8: Template 2: GENERAL INTERNAL SECURITY — Fight Against ____

FILL-IN-THE-BLANK INTRO

- "The fight against ____ is not that of any one govt/agency but requires all nations & stakeholders to come on the same platform and make coordinated efforts with equal intensity & seriousness."

FILL-IN-THE-BLANK CONCLUSION

- "If implemented well, ____ step should go a long way in tackling the issue of ____."